SolarLab Technical and User Manual

Technical guide to thin-film solar-cell simulation, workflows, and validation

SolarLab Project

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Chapter 1

Executive Summary

SolarLab is a thin-film photovoltaic device simulator that couples drift-diffusion transport, Poisson electrostatics, mobile-ion redistribution, recombination, and optical generation within a reproducible software workflow. The numerical core is exposed through a Python API, a FastAPI service layer, and a TypeScript/Plotly workstation. An experimental two-dimensional extension supports lateral microstructure and grain-boundary studies.

This manual serves two purposes:

1. It introduces the device variables, parameter conventions, and simulation workflow for readers who are new to SolarLab or to semiconductor device simulation.
2. It documents the implemented physics, numerical method, validation gates, assumptions, and limitations needed for technical review and reproducible use.

The software validation status — the unit, regression, and physics-trend test suites, the benchmark envelopes, and the repository state they certify — is reported in Chapter 16; the corresponding model assumptions and limitations are collected in the chapters that follow it.

Chapter 2

How To Read This Manual

If you are new to device simulation, read the chapters in order through *Device Definition*, *Governing Equations*, and *Numerical Method*. These chapters explain the device, the variables, the equations, and the numerical method before introducing the software workflow.

If you are already familiar with semiconductor drift-diffusion, start with:

- Chapter 7 (*Governing Equations*), which defines the physical model in continuous form;
- Chapter 10 (*Numerical Method*), which describes the discretization, the time integrator, and the steady-state solver;
- Chapter 11 (*Running SolarLab*), which covers the YAML/Python/API workflows;
- Chapter 16 (*Validation And Evidence*), which gives validation evidence and limitations.

The notation in this manual follows standard solar-cell convention:

- V_{oc}: open-circuit voltage.
- J_{sc}: short-circuit current density.
- FF: fill factor.
- PCE: power-conversion efficiency.
- n: electron density.
- p: hole density.
- P: mobile positive ion-vacancy density.
- ϕ: electrostatic potential.
- E: electric field.
- G: optical generation rate.
- R: recombination rate.

All simulator inputs are SI unless explicitly stated otherwise. Length is in meters, time in seconds, current density in $A\,m^{-2}$, density in m^{-3}, mobility in $m^2V^{-1}s^{-1}$, and diffusion coefficient in m^2s^{-1}. The electron affinity χ and band gap E_g are stored in eV.

Chapter 3

Simulator Architecture

SolarLab should be read as a coupled model implementation rather than as a stand-alone user interface. YAML files, the Python API, the FastAPI backend, and the TypeScript frontend all operate on the same device schema. A parameter edited in the browser is therefore serialized into the same scientific object used by the solver.

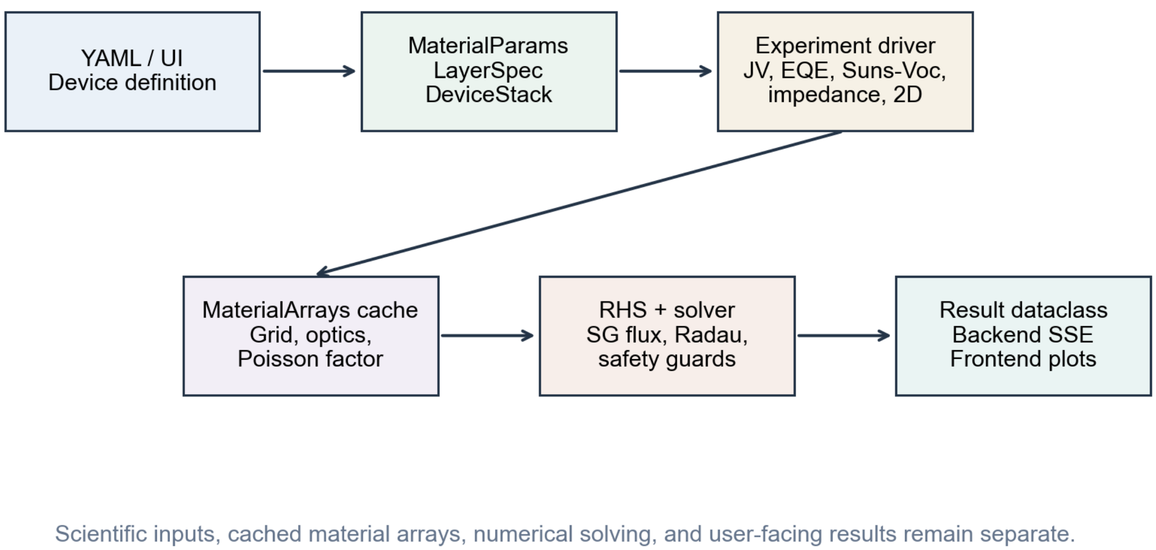

Figure 3.1: SolarLab architecture flow

The architecture preserves four technical boundaries:

- device definition is separated from experiment settings;
- material and grid arrays are cached before the time integrator runs;
- physics hooks are activated by the resolved simulation mode and by the presence of required parameters;
- backend and frontend layers transport result dataclasses instead of reimplementing solver logic.

For reproducible interpretation, each reported curve should identify the device stack, physics tier, experiment driver, and solver configuration. Missing metadata weakens reproducibility even when the numerical run completes successfully.

Chapter 4

What SolarLab Simulates

SolarLab represents a solar cell as a sequence of material layers between two contacts. The default coordinate follows the stack direction. Incident light generates electron-hole pairs, while the electrostatic field and contact selectivity drive carrier separation and extraction. In perovskite devices, mobile ionic defects can redistribute under the same field, producing history-dependent current-voltage response.

The default solver is one-dimensional and is appropriate for:

- J-V sweeps and hysteresis;
- dark diode curves;
- EQE and electroluminescence;
- impedance;
- transient photovoltage;
- ion-coupled degradation;
- tandem current matching;
- screening many material candidates.

SolarLab also includes an experimental two-dimensional solver. The 2D solver extrudes the same vertical device stack laterally and is intended for:

- 1D/2D parity checks;
- vertical grain boundaries;
- lateral microstructure;
- $V_{\text{oc}}(L_g)$ grain-size sweeps.

The 2D solver is not a replacement for the 1D workflow. Mobile ions are held as a static Poisson background during 2D J-V runs; this is an explicit model assumption and should be reported when interpreting 2D results.

Illumination is always incident at $x = 0$, through the first listed electrical layer (after any optical-only substrate layers). The device-structure figure shows the layer order of the shipped `nip_MAPbI3` preset — HTL at the illuminated front face, then absorber, then ETL — so the coordinate direction in the figure matches the YAML layer order exactly. Note that this ordering does not follow the common community reading of "n-i-p" as light entering through the n-side transport layer: in SolarLab, device orientation is defined *solely* by the YAML layer order together with the $x = 0$ illumination convention, and preset names should be read as stack identifiers, not as a statement of which side faces the sun. Users porting structures from the literature should verify the intended illumination side against the layer order.

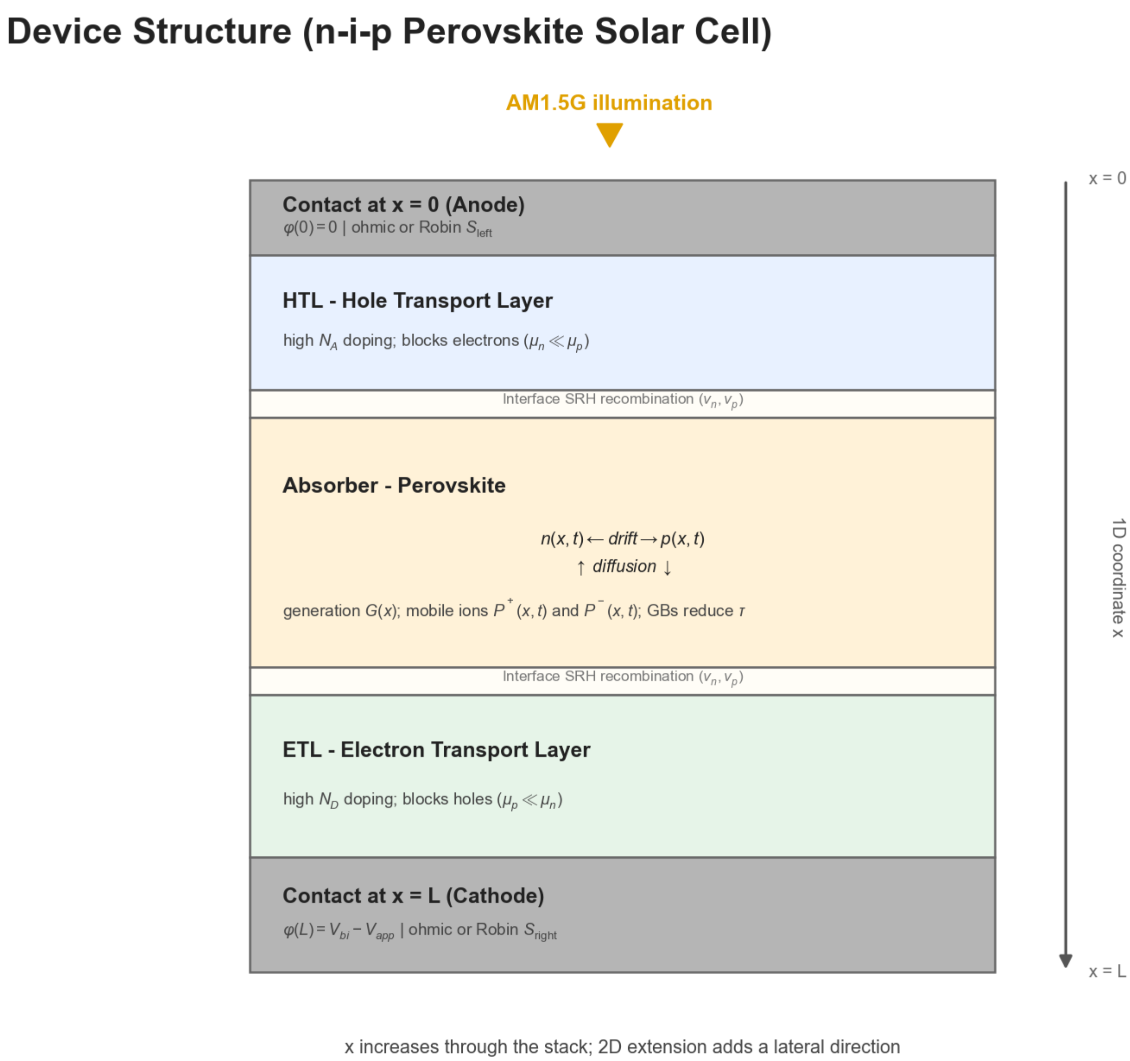

Figure 4.1: SolarLab device structure

Energy Band Diagram

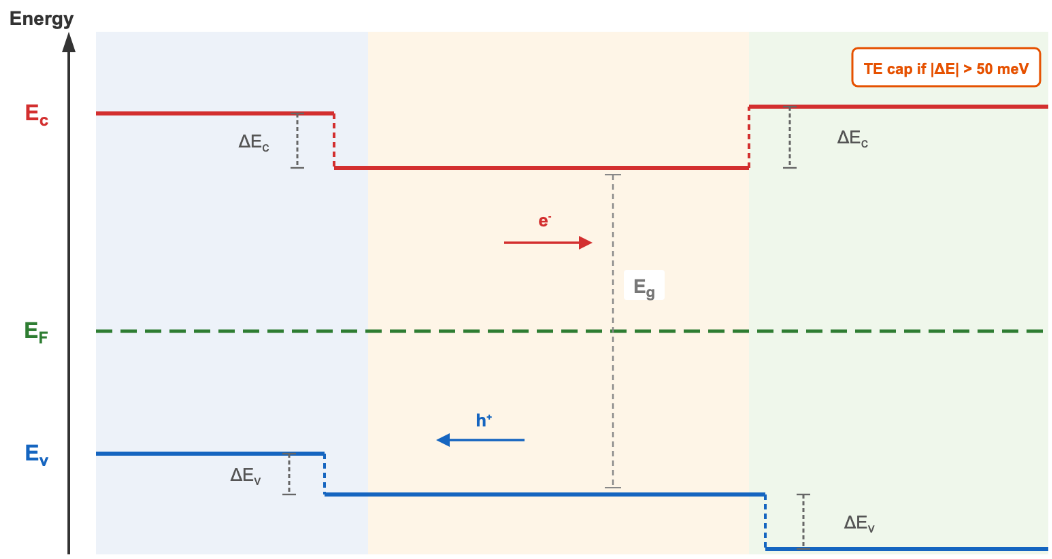

Figure 4.2: SolarLab band-diagram concept

Chapter 5

Beginner Physics Primer

5.1 Electrons, Holes, And Bands

In a semiconductor, electrons can move in the conduction band and holes can move in the valence band. A solar cell works because light creates electron-hole pairs and the device structure makes it more likely that electrons leave through one contact while holes leave through the other.

SolarLab stores two energy parameters that are essential for heterojunctions:

- χ, the electron affinity;
- E_g, the band gap.

Together these determine approximate conduction-band and valence-band positions. At an interface, a change in χ or E_g creates band offsets. Band offsets are important because they can block one carrier type while allowing the other carrier type to pass.

5.2 Generation And Recombination

Optical generation $G(x)$ is the rate at which absorbed photons create electron-hole pairs per unit volume. Recombination $R(n, p)$ is the rate at which electrons and holes annihilate. A good solar cell has high generation and carrier collection, but low recombination.

SolarLab includes several recombination channels:

- Shockley-Read-Hall (SRH) trap-assisted recombination;
- radiative recombination;
- Auger recombination;
- interface recombination through surface recombination velocities;
- optional trap-density profiles near interfaces.

5.3 Mobile Ions And Hysteresis

Perovskite cells can contain mobile ionic defects. These defects move much more slowly than electronic carriers. During a voltage scan, the ion profile may not reach equilibrium at each voltage. This creates J-V hysteresis. SolarLab treats this as a physical transient, not as a post-processing correction.

In the state vector, the default mobile ion variable is P, the positive vacancy density. A second negative species can be configured through `D_ion_neg`, `P0_neg`, and `P_lim_neg`.

Chapter 6

Device Definition

6.1 Core Data Flow

The simulator data flow is:

```
YAML or inline device dictionary
-> MaterialParams + LayerSpec + DeviceStack
-> experiment driver
-> MaterialArrays cache
-> result dataclass
```

The main dataclasses are:

- `MaterialParams`: physical parameters for one material.
- `LayerSpec`: layer name, role, thickness, and material parameters.
- `DeviceStack`: full multilayer stack and global device settings.

6.2 Layer Roles

Layer roles used by the frontend and YAML schema are:

Role	Meaning
`substrate`	optical-only layer, excluded from electrical drift-diffusion
`front_contact`	front electrode or transparent conductor
`ETL`	electron transport layer
`absorber`	active photovoltaic absorber
`HTL`	hole transport layer
`back_contact`	rear electrode

Substrate layers must form a contiguous prefix of the stack. They participate in transfer-matrix optics but do not appear in the drift-diffusion grid.

6.3 Main Material Parameters

Field	Meaning	Unit
`eps_r`	relative permittivity	dimensionless
`mu_n`, `mu_p`	electron and hole mobilities	$m^2V^{-1}s^{-1}$
`D_ion`	positive ion diffusion coefficient	m^2s^{-1}
`P_lim`	positive ion steric limit	m^{-3}
`P0`	initial positive ion density	m^{-3}
`ni`	intrinsic carrier density	m^{-3}
`tau_n`, `tau_p`	SRH lifetimes	s
`n1`, `p1`	SRH trap reference densities	m^{-3}
`B_rad`	radiative recombination coefficient	m^3s^{-1}
`C_n`, `C_p`	Auger coefficients	m^6s^{-1}
`alpha`	Beer-Lambert absorption coefficient	m^{-1}
`N_A`, `N_D`	acceptor and donor densities	m^{-3}
`chi`	electron affinity	eV
`Eg`	band gap	eV
`optical_material`	key into `data/nk/*.csv`	string
`incoherent`	incoherent TMM layer flag	boolean

Additional optional parameters cover negative ions, temperature scaling, field-dependent mobility, trap profiles, and optical fallbacks.

6.4 Parameter Dictionary

Input specification is a dominant source of simulation error. SolarLab cannot infer whether a parameter comes from measurement, literature, fitting, or exploratory screening, so each field should be recorded with its source and value.

6.4.1 Electrical And Transport Fields

Field	Unit	Physical role	Beginner guidance
`eps_r`	1	relative permittivity in Poisson's equation	Larger values screen charge and reduce field gradients. Use layer-specific literature values rather than a single generic perovskite value.
`mu_n`	$m^2V^{-1}s^{-1}$	low-field electron mobility	Controls electron extraction and transport resistance. Remember that $1\,cm^2V^{-1}s^{-1} = 10^{-4}\,m^2V^{-1}s^{-1}$.
`mu_p`	$m^2V^{-1}s^{-1}$	low-field hole mobility	Same unit conversion convention as `mu_n`. Low mobility typically affects FF before producing a large change in V_{oc}.
`N_A`	m^{-3}	ionized acceptor density	Represents p-type doping. Values reported in cm^{-3} must be multiplied by 10^6.
`N_D`	m^{-3}	ionized donor density	Represents n-type doping. Use compensated doping only when it is part of the intended physical model.

Field	Unit	Physical role	Beginner guidance
ni	m^{-3}	intrinsic carrier density	Appears in mass action and SRH terms. It should be consistent with the effective band gap and temperature when doing parameter studies.
chi	eV	electron affinity	Sets approximate conduction-band alignment. Inconsistent χ values can introduce artificial transport barriers.
Eg	eV	band gap	Sets band offsets and thermodynamic interpretation. Beer-Lambert optics do not automatically shift spectral absorption when Eg changes.

6.4.2 Ion And Hysteresis Fields

Field	Unit	Physical role	Beginner guidance
D_ion	$m^2 s^{-1}$	positive mobile-ion diffusion coefficient	Set to zero when mobile ions are excluded from the model. Nonzero values can produce scan-rate dependence.
P0	m^{-3}	initial positive ion density	Represents the equilibrium mobile-vacancy population before redistribution.
P_lim	m^{-3}	steric upper density for positive ions	Limits ion accumulation through a finite-site-density approximation. The value should be consistent with the assumed mobile-site density.
D_ion_neg	$m^2 s^{-1}$	negative mobile-ion diffusion coefficient	Enables a second mobile species. Leave zero for the default single-ion model.
P0_neg	m^{-3}	initial negative ion density	Use only when the negative species is part of the physical hypothesis.
P_lim_neg	m^{-3}	steric upper density for negative ions	Same interpretation as P_lim.
E_a_ion	eV	Arrhenius activation energy for ion diffusion	Used for temperature-dependent ion mobility. Fitted or literature-derived values should be reported with provenance.

6.4.3 Recombination And Trap Fields

Field	Unit	Physical role	Beginner guidance
tau_n	s	electron SRH lifetime	Shorter lifetime increases trap-assisted recombination. It is often an effective fitted parameter.
tau_p	s	hole SRH lifetime	Interpret with tau_n; asymmetry can model carrier-selective trap response.
n1	m^{-3}	SRH electron reference density	Encodes the effective trap-energy position and should be changed consistently with the trap model.
p1	m^{-3}	SRH hole reference density	Companion to n1.

Field	Unit	Physical role	Beginner guidance
`B_rad`	$m^3 s^{-1}$	radiative recombination coefficient	Central to radiative-limit and photon-recycling studies.
`C_n`	$m^6 s^{-1}$	electron Auger coefficient	Most relevant when high carrier densities are expected.
`C_p`	$m^6 s^{-1}$	hole Auger coefficient	Same caution as `C_n`.
`trap_N_t_interface`	m^{-3}	interface-near trap density	Activates spatial trap profiles when supplied with decay information.
`trap_N_t_bulk`	m^{-3}	bulk trap density	The asymptotic trap density away from the interface.
`trap_decay_length`	m	decay length or Gaussian width	Must be physically resolvable by the grid.
`trap_profile_shape`	string	`exponential` or `gaussian` trap profile	Select the profile shape that corresponds to the assumed defect distribution.

6.4.4 Optical Fields

Field	Unit	Physical role	Beginner guidance
`alpha`	m^{-1}	scalar Beer-Lambert absorption coefficient	Suitable for simplified optical studies. It is not wavelength-resolved.
`optical_material`	string	key into tabulated n, k data	Required for TMM, EQE, and EL. The key must exist in `perovskite_sim/data/nk`.
`n_optical`	1	constant refractive-index fallback	Useful for approximate optical calculations but not a substitute for measured n, k.
`incoherent`	boolean	thick-layer incoherent TMM treatment	Intended for the first thick substrate. The current TMM path does not support arbitrary incoherent layers mid-stack.

6.4.5 Temperature And Field-Dependent Mobility Fields

Field	Unit	Physical role	Beginner guidance
`Nc300`	m^{-3}	conduction-band density of states at 300 K	Optional temperature-scaling input. Use only when the temperature model is calibrated.
`Nv300`	m^{-3}	valence-band density of states at 300 K	Companion to `Nc300`.
`mu_T_gamma`	1	mobility power-law temperature exponent	Default keeps a common scattering-style trend. Document any changed value.
`B_rad_T_gamma`	1	radiative coefficient temperature exponent	Default zero preserves legacy behavior.
`varshni_alpha`	eV K^{-1}	Varshni band-gap coefficient	Zero disables band-gap temperature shift.
`varshni_beta`	K	Varshni temperature parameter	Used with `varshni_alpha`.
`v_sat_n`	$m\,s^{-1}$	electron velocity-saturation limit	Zero disables Caughey-Thomas saturation for electrons.

Field	Unit	Physical role	Beginner guidance
`v_sat_p`	$m\,s^{-1}$	hole velocity-saturation limit	Zero disables Caughey-Thomas saturation for holes.
`ct_beta_n`	1	electron Caughey-Thomas exponent	Controls sharpness of velocity saturation.
`ct_beta_p`	1	hole Caughey-Thomas exponent	Same role for holes.
`pf_gamma_n`	$(V/m)^{-1/2}$	electron Poole-Frenkel mobility coefficient	Zero disables field-enhanced hopping for electrons.
`pf_gamma_p`	$(V/m)^{-1/2}$	hole Poole-Frenkel mobility coefficient	Zero disables field-enhanced hopping for holes.

6.4.6 Device-Level Fields

Field	Unit	Physical role	Beginner guidance
`V_bi`	V	built-in voltage in Poisson boundary condition	This is not always identical to the derived heterostack Fermi-level separation.
`Phi`	$m^{-2}s^{-1}$	incident photon flux for Beer-Lambert generation	For spectral experiments use TMM data rather than only changing `Phi`.
`T`	K	device temperature	Affects thermal voltage and enabled temperature-dependent hooks.
`mode`	string	`legacy`, `fast`, or `full` physics tier	The mode is a ceiling; hooks still need required parameters.
`interfaces`	$m\,s^{-1}$ pairs	interface recombination velocities	One pair per internal electrical interface, ordered from the front contact toward the rear contact.
`S_n_left`	$m\,s^{-1}$	left electron contact velocity	`None` gives ohmic default. `0` is blocking. Large values approach ohmic extraction.
`S_p_left`	$m\,s^{-1}$	left hole contact velocity	Same convention as `S_n_left`.
`S_n_right`	$m\,s^{-1}$	right electron contact velocity	Same convention as `S_n_left`.
`S_p_right`	$m\,s^{-1}$	right hole contact velocity	Same convention as `S_n_left`.
`microstructure`	object	2D grain-boundary specification	Used by 2D drivers; ignored by standard 1D paths.

6.5 Global Device Parameters

`DeviceStack` stores:

Field	Meaning
`V_bi`	built-in voltage used in the Poisson boundary condition
`Phi`	incident photon flux for Beer-Lambert generation
`interfaces`	interface recombination velocities (v_n, v_p)
`T`	device temperature in K
`mode`	`legacy`, `fast`, or `full`
`S_n_left`, `S_p_left`, `S_n_right`, `S_p_right`	selective-contact velocities
`microstructure`	optional 2D grain-boundary block

6.6 Built-In Potential

SolarLab distinguishes two built-in potentials that follow two different sign conventions. Keeping them separate is deliberate.

1. `stack.V_bi` — the value read from YAML. It is a *positive built-in-potential magnitude* and is the value the **default** Poisson boundary condition uses. This is the IonMonger convention, in which V_{bi} is treated as a free parameter representing the degenerate-doping (flat-band-metal) limit rather than a quantity derived from the band structure. The orientation of the internal field is carried by the doping-derived contact densities, so the boundary value itself is supplied as a positive magnitude.

2. `stack.compute_V_bi()` — a *derived, signed* built-in potential equal to the contact work-function difference,

$$V_{\text{bi}}^{\text{eff}} = \phi(L) - \phi(0) = W_{\text{left}} - W_{\text{right}},$$

obtained from the equilibrium Fermi-level separation across the electrical stack (χ, E_g, doping, and n_i). Because it is the signed contact potential, it is *positive for the usual p-contact-left orientation and negative for an n-contact-left device* (for example an n-window / p-absorber CIGS stack, or an n^+/p silicon homojunction). The negative sign is physically correct: the n-side contact holds the positive donor space charge and equilibrates at the higher electrostatic potential. When every electrical layer has $\chi = E_g = 0$ (legacy configurations with no band data), `compute_V_bi()` falls back to the manual `stack.V_bi`.

The two therefore need not agree even in magnitude — a benchmark may pin a free-parameter V_{bi} that differs from the band-derived value (the IonMonger reference, for instance, uses 1.10 V where the Fermi-level construction gives 0.86 V).

The Poisson boundary is:

$$\phi(0) = \phi_{\text{left}}, \qquad \phi(L) = \phi_{\text{left}} + V_{\text{bi}} - V_{\text{app}}.$$

By default the boundary uses the manual `stack.V_bi` (item 1). Setting `flat_band_contacts: true` switches it to the derived, signed `compute_V_bi()` (item 2), reproducing the SCAPS flat-band-metal contact convention. The derived value is also used — as a magnitude, via $|\cdot|$ — for solver defaults such as the automatic upper voltage of a J-V sweep and the initial bracket of an open-circuit search.

6.7 YAML Example

```yaml
device:
  V_bi: 1.1              # manual built-in-potential magnitude (default Poisson
                         # BC, IonMonger convention). Set flat_band_contacts to
                         # derive the signed value from the bands instead.

  Phi: 2.5e21
  T: 300.0
  mode: full
  contacts:
    left:
      S_p: 1.0e3
```

```yaml
      S_n: 1.0e-3
    right:
      S_n: 1.0e3
      S_p: 1.0e-3

layers:
  - name: HTL
    role: HTL
    thickness: 2.0e-7
    eps_r: 3.0
    mu_n: 1.0e-8
    mu_p: 1.0e-6
    ni: 1.0e9
    N_A: 1.0e24
    N_D: 0.0
    D_ion: 0.0
    P_lim: 1.0e27
    P0: 0.0
    tau_n: 1.0e-6
    tau_p: 1.0e-6
    n1: 1.0e9
    p1: 1.0e9
    B_rad: 0.0
    C_n: 0.0
    C_p: 0.0
    alpha: 0.0
    chi: 2.2
    Eg: 3.0
```

YAML 1.1 can parse bare scientific notation such as `1e-9` as a string. SolarLab coerces numeric-looking strings to floats, but writing values as `1.0e-9` is clearer and safer.

Chapter 7

Governing Equations

This chapter states the governing equations in dimensional SI form. The model belongs to the class of coupled Poisson/drift-diffusion (van Roosbroeck) systems, extended by Nernst-Planck transport for mobile ionic defects, heterojunction band alignment, and wavelength-resolved optical generation. The implementation evaluates discretized node and face arrays (Chapter 10), but the continuous form stated here clarifies the model assumptions and the physical meaning of each state variable.

Throughout, q is the elementary charge, k_B Boltzmann's constant, T the lattice temperature, and $V_T = k_B T/q$ the thermal voltage. Carrier statistics are Maxwell-Boltzmann (non-degenerate); Fermi-Dirac corrections are outside the present scope and matter only for degenerately doped layers.

7.1 Electrostatics: The Poisson Equation

The electrostatic potential $\phi(x,t)$ is obtained from the quasistatic Poisson equation

$$\frac{\partial}{\partial x}\left(\varepsilon_0\varepsilon_r\frac{\partial\phi}{\partial x}\right) = -\rho, \tag{7.1}$$

with the space-charge density assembled from every charged species in the model:

$$\rho = q\left[p - n + (P - P_0) - (P^- - P_0^-) + N_D - N_A\right]. \tag{7.2}$$

The mobile-ion contributions are measured relative to their neutral background densities P_0 and P_0^-: an undisturbed vacancy population together with its immobile counter-charge is electrically neutral, so only the *redistribution* of ions contributes net space charge. The negative-species term is present only when a second mobile species is configured (dual-ion mode).

The quasistatic approximation asserts that the potential responds instantaneously to the charge configuration — Eq. 7.1 is Gauss's law $\nabla\cdot(\varepsilon_0\varepsilon_r\mathbf{E}) = \rho$ with $E = -\partial\phi/\partial x$, solved at every instant of the transient. Retardation, magnetic effects, and electromagnetic wave propagation are excluded from the electrical solve (the optical field is treated separately by the transfer-matrix machinery of the *Optical Generation* section).

The discretization is finite-volume with a harmonic-mean face permittivity:

$$\varepsilon_{i+1/2} = \frac{2\varepsilon_i \varepsilon_{i+1}}{\varepsilon_i + \varepsilon_{i+1}}. \tag{7.3}$$

When the material interface falls at the midpoint between the two nodes, the harmonic mean is the exact series composition of the two dielectric half-cells: it is the face permittivity for which the discrete flux reproduces a continuous normal displacement $D = \varepsilon_0 \varepsilon_r E$ across the discontinuity, with the field jumping between the layers as it physically must. An arithmetic (nodal) mean is the *parallel* composition instead, which is the correct average for a face-tangential field, so using it here misallocates the potential drop between the two layers. Neither choice literally imposes a boundary condition on E; the difference is which equivalent series-versus-parallel dielectric response the face flux represents.

For a general non-symmetric geometry, where the interface does not sit at the midpoint, the correct series composition is the *distance-weighted* harmonic mean

$$\varepsilon_f = \frac{\Delta x_L + \Delta x_R}{\Delta x_L/\varepsilon_L + \Delta x_R/\varepsilon_R},$$

which reduces to Eq. 7.3 for equal half-widths. SolarLab places every material interface *on* a grid node and assigns that node the permittivity of the layer to its right, so each face lies entirely within one material and Eq. 7.3 smears the discontinuity over one half-cell rather than weighting two dissimilar half-cells. This is the usual node-centred finite-volume treatment and its error is first order in the local cell width; it is *not* a distance-weighting bug, but it does mean the interface is resolved to within half a cell, so interface-sensitive quantities should be checked for mesh convergence.

Assumptions and limits:

- electrostatics is quasistatic;
- magnetic effects and wave propagation in the electrical solve are ignored;
- layer interfaces are abrupt on the scale of the grid;
- ρ is built from the configured mobile species, carriers, and dopants, with dopants assumed fully ionized and immobile.

7.2 Carrier Transport: Continuity And Drift-Diffusion

The electron and hole densities obey the continuity equations

$$\frac{\partial n}{\partial t} = \frac{1}{q}\frac{\partial J_n}{\partial x} + G - R, \tag{7.4}$$

$$\frac{\partial p}{\partial t} = -\frac{1}{q}\frac{\partial J_p}{\partial x} + G - R, \tag{7.5}$$

closed by the drift-diffusion constitutive relations

$$J_n = q\mu_n n E + qD_n \frac{\partial n}{\partial x}, \qquad J_p = q\mu_p p E - qD_p \frac{\partial p}{\partial x}, \tag{7.6}$$

with the Einstein relation $D = \mu V_T$ linking diffusivity and mobility. The Einstein relation expresses local thermal equilibrium between the carrier gas and the lattice; it is exact for non-degenerate (Boltzmann) statistics.

An equivalent and physically transparent form writes each current as a quasi-Fermi-level gradient,

$$J_n = \mu_n \, n \, \frac{\partial E_{Fn}}{\partial x}, \qquad J_p = \mu_p \, p \, \frac{\partial E_{Fp}}{\partial x}, \tag{7.7}$$

where, under Boltzmann statistics, $n = N_C \exp[-(E_C - E_{Fn})/k_B T]$ and $p = N_V \exp[-(E_{Fp} - E_V)/k_B T]$. Equations 7.6 and 7.7 are the same model; the quasi-Fermi form makes explicit that a flat quasi-Fermi level implies zero current *for that carrier* — the thermodynamic-equilibrium condition. At open circuit only the net terminal current is constrained to zero: internal electron and hole currents remain individually non-zero and convert into each other through generation and recombination, so the quasi-Fermi levels need not be flat everywhere in an operating device.

7.2.1 Heterojunction Band Alignment

Band edges are referenced to the local vacuum level through the electron affinity χ and the band gap E_g:

$$E_C/q = -(\phi + \chi), \qquad E_V/q = -(\phi + \chi + E_g),$$

with χ and E_g expressed in volts (numerically equal to their eV values). In heterojunction stacks SolarLab therefore drives the carrier fluxes with *band-effective* potentials,

$$\phi_n = \phi + \chi, \qquad \phi_p = \phi + \chi + E_g, \tag{7.8}$$

so that electrons respond to gradients of the conduction band edge and holes to gradients of the valence band edge. An abrupt change in χ or E_g between layers then produces Anderson-rule band offsets (ΔE_C from the affinity step, ΔE_V from the combined affinity-plus-gap step) inside the same numerical flux formulation, with no special-case interface treatment in the discretization.

7.2.2 Effective-Density-of-States Correction

For a heterostructure with unequal effective densities of states, the band-effective potentials of Eq. 7.8 are incomplete. Under Boltzmann statistics the carrier density couples to the quasi-Fermi level through the local N_C or N_V, so the exact drift potentials carry an additional logarithmic term:

$$\phi_n = \phi + \chi + V_T \ln \frac{N_C}{N_C^{\text{ref}}}, \qquad \phi_p = \phi + \chi + E_g - V_T \ln \frac{N_V}{N_V^{\text{ref}}}. \tag{7.9}$$

Omitting these terms imposes a spurious quasi-Fermi-level discontinuity of $k_B T \ln(N_{C,1}/N_{C,2})$ (and the N_V analogue) at every interface with a density-of-states contrast — on a representative HTL/perovskite/ETL stack the accumulated error exceeds 100 mV of open-circuit voltage. SolarLab therefore folds the corrections of Eq. 7.9 into the cached per-layer χ/E_g arrays at build time, using the absorber as the reference layer (only cross-junction *differences* matter). The fold applies to transport and thermionic emission only; purely statistical quantities (n_i, the SRH references n_1/p_1, and the contact equilibrium densities) are left unchanged. The correction is active by default whenever per-layer `Nc300`/`Nv300` data are supplied, and is disabled in the `legacy` tier, which reproduces the IonMonger convention without DOS-folded transport.

Assumptions and limits:

- carriers are described by drift-diffusion transport, not ballistic transport;
- carrier statistics are non-degenerate (Maxwell-Boltzmann); degenerately doped layers would require Fermi-Dirac corrections that are out of scope;
- the model is most natural when local quasi-equilibrium is a reasonable approximation;
- strongly quantum-confined structures, tunneling-dominated contacts, and hot carrier effects require additional modeling beyond this implementation.

7.3 Heterointerface Thermionic Emission And Tunnelling

The drift-diffusion picture assumes that the band edges vary smoothly on the scale of a grid cell. At an abrupt heterojunction the band offset is instead resolved within a *single* cell. With the band offsets and density-of-states corrections carried in the exponentially fitted potentials, the Scharfetter-Gummel flux (Chapter 10) is then the ideal-interface limit: it assumes local equilibrium across the step and unlimited interface transmission, i.e. no additional kinetic resistance at the junction. That is an upper bound on the transport a real interface supports whenever interface scattering, a finite emission velocity, tunnelling, or interface-state supply actually limit it — it is not a systematic discretization error for every band step, and the size of the discrepancy is a property of the interface, not of the scheme. SolarLab therefore bounds the drift-diffusion face flux by an interface-limited thermionic-emission (TE) flux of Richardson-Dushman form at every face where the conduction- or valence-band offset exceeds 50 meV:

$$J_{\mathsf{TE}} \;\propto\; A^{*}T^{2}\left[n_L\, e^{-\,\mathsf{max}(\Delta E,\,0)/V_T} - n_R\, e^{-\,\mathsf{max}(-\Delta E,\,0)/V_T}\right], \tag{7.10}$$

where ΔE is the band-edge step across the face, n_L and n_R are the adjacent carrier densities, and A^* is a per-face Richardson constant (default: the free-electron value $1.2017 \times 10^6\, A\, m^{-2} K^{-2}$, overridable per layer). Only the *uphill* term carries the Boltzmann barrier penalty; a downhill offset leaves the emission term unpenalized, so carriers flowing down a band step are not artificially blocked.

The default form carries two qualifications. First, the density-weighted form of Eq. 7.10 is an *empirical interface-limited bound*, not the dimensional Richardson-Dushman current: A^*T^2 is already a current density, so multiplying by an absolute carrier density leaves the bound's magnitude tied to the SI unit system. This makes $|J_{TE}|$ very large (order 10^{28}–10^{35} on a perovskite stack), so the smaller-magnitude cap is rarely active; the observed heterojunction V_{oc} behaviour is driven by the band-offset potentials of Eqs. 7.8–7.9, not by the TE cap. A dimensionally correct *emission-velocity* form is available as an opt-in (`te_physical_norm`): dividing Eq. 7.10 by the band-edge density of states converts A^*T^2 into an emission velocity $v_R = A^*T^2/(qN_C)$, giving $J = qv_R(\dots)$, which binds at real interface densities. Enabling it changes the physics measurably where it acts: on a density-of-states-bearing stack the steady-state V_{oc} rises by roughly 0.3 V once the cap begins to bind. This hook acts only on the bulk-node transport cap (used by the transient driver and the direct steady-state solver); the interface-plane-state formulation employed for the SCAPS comparison carries its own thermionic fluxes and is *unaffected* by the flag (measured V_{oc} identical to four decimals with it on or off), so the calibrated parity result is untouched. It is kept off by default because the paths it does affect have no independent reference baseline with the physical cap active (Chapter 17); configurations without per-layer effective-DOS data are bit-identical whether it is on or off. Second, the two-leg bracket itself vanishes at thermodynamic equilibrium only when the adjacent layers share the same effective density of states or when the density-of-states-folded potentials of Eq. 7.9 are active. System-level equilibrium is nevertheless preserved unconditionally by the *cap construction*: at equilibrium the drift-diffusion flux is exactly zero, and the smaller-magnitude selection below therefore returns zero regardless of the value of J_{TE}.

The TE flux acts as a *cap*: at each flagged face the solver takes the smaller-magnitude of the drift-diffusion flux and J_{TE}, so the interface-limited current can only reduce, never enhance, the drift-diffusion prediction. (The steady-state Newton driver replaces the hard minimum with a smooth sigmoid blend of adjustable width to restore differentiability; see Chapter 10.)

7.3.1 Thermionic-Field-Emission Tunnelling

Pure thermionic emission transports carriers *over* the barrier only. An optional intra-band tunnelling channel augments it with transport *through* the top of a band spike, as a **doping-derived static tunnelling enhancement** scaled by the Padovani-Stratton characteristic energy E_{00}. It is not an implementation of the Padovani-Stratton thermionic-field-emission current: that theory evaluates a current-voltage relation from an energy integral over a specific barrier shape, whereas the enhancement below is a single bias-independent factor derived from the doping. In particular it does not solve for the bias-dependent depletion width, the local field, or the barrier profile, and it is therefore not equation-level equivalent to the WKB tunnelling model in SCAPS (which computes a transmission probability with a numerical floor on the barrier height and enters the DC solution only). Cross-simulator comparisons should either disable tunnelling on both sides or benchmark the two channels separately against a common barrier and effective mass. The characteristic tunnelling energy is

$$E_{00} = \frac{\hbar}{2}\sqrt{\frac{N_{\mathsf{if}}}{m^* \varepsilon_s}}, \qquad E_0 = E_{00} \coth\left(\frac{E_{00}}{V_T}\right), \tag{7.11}$$

where E_{00} is expressed in eV (the prefactor $\hbar/2$, without the charge factor, yields volts directly; the implementation computes the joule value $(q\hbar/2)\sqrt{N_{\mathsf{if}}/(m^*\varepsilon_s)}$ and divides by q, which is the same number), so the ratio E_{00}/V_T in the coth is dimensionally consistent. N_{if} is the net doping of the lighter-doped (depletion) side of the junction, m^* the effective tunnelling mass (default $0.2\,m_e$), and ε_s the semiconductor permittivity. The tunnelling-transparent fraction of the barrier, $\delta_{\mathsf{tun}} = |\Delta E|\,(1 - V_T/E_0)$, clamped to $[0, |\Delta E|)$, defines a static enhancement factor

$$\Gamma = \exp\left(\frac{\delta_{\mathsf{tun}}}{V_T}\right) \geq 1, \tag{7.12}$$

which is folded into the per-face Richardson constants ($A^*_{\mathsf{eff}} = \Gamma A^*$) at build time. Because the same Γ scales *both* legs of Eq. 7.10, equilibrium detailed balance is preserved to machine precision, and because the factor is static (doping-derived, no dependence on the evolving field) it cannot perturb the stability of the time integration. The channel is opt-in and captures the doping-direction trend of tunnelling-assisted interface transport; the bias-dependent flattening of a spike under forward bias would require a field-dependent $E_{00}(F)$ evaluated per step, which is outside the present scope.

Assumptions and limits:

- TE capping applies only at faces whose band offset exceeds 50 meV; smaller offsets are handled by the drift-diffusion flux alone;

- the tunnelling enhancement is static and doping-derived; an intrinsic-absorber interface ($N_{\text{if}} \to 0$) gives $E_{00} \to 0$ and hence $\Gamma \to 1$ (no enhancement);
- the `legacy` tier disables thermionic emission entirely (IonMonger has no TE), and with it the tunnelling channel by construction.

7.4 Ion Migration

Mobile ionic defects — in halide perovskites, predominantly halide vacancies acting as an effective positive species — are transported by a Nernst-Planck equation of the same drift-diffusion form as the electronic carriers. The positive vacancy density P satisfies the conservation law

$$\frac{\partial P}{\partial t} = -\frac{\partial F_P}{\partial x}, \tag{7.13}$$

with the flux (for a $+1$ charge, using the Einstein relation for the ionic mobility)

$$F_P = -\,s(P)\,D_I\left(\frac{\partial P}{\partial x} + \frac{P}{V_T}\frac{\partial \phi}{\partial x}\right). \tag{7.14}$$

There is no generation or recombination term: the total number of mobile ions in the device is conserved. Consistently, the contacts are ion-blocking,

$$F_P(0) = F_P(L) = 0, \tag{7.15}$$

so that $\int_0^L P\,dx$ is an exact invariant of the dynamics — ions redistribute internally but never leave through the electrodes.

The prefactor $s(P)$ is a steric (excluded-volume) crowding correction motivated by modified Poisson-Nernst-Planck / lattice-gas theories:

$$s(P) = \frac{1}{\max(1 - P/P_{\text{lim}},\, 10^{-6})}. \tag{7.16}$$

As the local occupancy P approaches the finite site density P_{lim}, the prefactor diverges and the flux is sharply enhanced, which accelerates the relaxation of an interfacial pile-up relative to the ideal-solution model. (In the discretization, P in Eq. 7.16 is evaluated as the average of the two face-adjacent nodal densities; the 10^{-6} floor guards the division at full saturation.)

This default form does **not** impose $P \leq P_{\text{lim}}$, and should not be read as a finite-site-occupancy constraint. Because $s(P) > 0$ multiplies the whole flux, the zero-flux condition is unchanged,

$$\frac{\partial P}{\partial x} + \frac{P}{V_T}\frac{\partial \phi}{\partial x} = 0,$$

i.e. the equilibrium is still the ideal-solution Boltzmann distribution, which itself admits $P > P_{\text{lim}}$; a common positive factor rescales the kinetics, not the equilibrium occupancy. The 10^{-6} floor makes the coefficient large, but largeness is not a bound, and a finite time step can still cross the limit. A true finite-occupancy model requires the lattice-gas chemical potential $\mu/k_BT = \ln[\theta/(1-\theta)] + q\phi/k_BT$ with $\theta = P/P_{\text{lim}}$, which is what the opt-in form below implements.

By default SolarLab applies $s(P)$ to the *entire* flux of Eq. 7.14, scaling diffusion and drift together; the strict lattice-gas chemical-potential derivation enhances only the concentration-gradient term. The dimensionally faithful form is available as an opt-in (`ion_steric_diffusion_only`): it folds the crowding chemical potential $\mu_{\text{ex}}/k_BT = -\ln(1 - P/P_{\text{lim}})$ into the Scharfetter-Gummel drift argument, so the excluded-volume correction acts on diffusion only while the exponential-fitting stability is preserved, and the equilibrium recovers the lattice-gas (Fermi-Dirac) occupation $\theta/(1-\theta) \propto e^{-\phi/V_T}$. The distinction is numerically immaterial for the shipped presets: the mobile-ion occupancy stays dilute (peak $P/P_{\text{lim}} \approx 0.011$ on the IonMonger benchmark, so $s \approx 1.011$), and enabling the physical form shifts V_{oc} by under 0.1 mV. The two forms diverge only as $P \to P_{\text{lim}}$ (extreme interfacial pile-up). For a dual mobile-ion system the diffusion-only form additionally couples the species through their occupancy: with `ion_steric_shared_site` (default) the crowding potential uses the total occupancy $1-(P_+ + P_-)/P_{\text{lim}}$, the standard multi-species finite-size treatment in which both species compete for one finite reservoir (it reduces exactly to the single-species form when one density vanishes); setting it false gives distinct sublattices, each species crowding only against its own P_{lim} — a physical assumption that should be stated explicitly for the device in question. The negative-species equation carries the same diffusion-only form symmetrically. The default whole-flux form is an empirical crowding regularization, not a finite-occupancy constraint (Chapter 17).

Because ionic diffusivities ($D_I \sim 10^{-16}\, m^2 s^{-1}$) are many orders of magnitude below carrier diffusivities, ionic redistribution introduces a slow time scale that coexists with nanosecond electronic relaxation. The governing relaxation is the drift-assisted screening of the built-in field across the interfacial, Debye-scale charge layers — typically milliseconds to seconds at perovskite ion mobilities — rather than diffusion over the full device length, which would be slower still. This stiff two-time-scale structure is the physical origin of scan-rate-dependent J-V hysteresis, and it is what the implicit time integration of Chapter 10 is designed to handle.

Additional model features:

- a second, negatively charged mobile species is optional and obeys the same equations with the drift sign reversed;
- ionic diffusivity is Arrhenius-activated in temperature through the per-layer activation energy `E_a_ion` when temperature scaling is enabled;
- non-ionic layers simply set $D_I = 0$; the ion equations still integrate but contribute zero flux.

Assumptions and limits:

- the default mobile species is a positive vacancy-like species;
- negative mobile species are optional and must be explicitly configured;
- contacts are ion-blocking in the implemented boundary condition;
- the 2D J-V solver treats ions as a frozen Poisson background.

7.5 Recombination

All bulk recombination channels share the mass-action driving term $np - n_i^2$, which vanishes at thermodynamic equilibrium and guarantees that no channel generates or destroys carriers in the dark equilibrium state.

7.5.1 Shockley-Read-Hall (Trap-Assisted) Recombination

$$R_{\text{SRH}} = \frac{np - n_i^2}{\tau_p(n + n_1) + \tau_n(p + p_1)}. \tag{7.17}$$

The reference densities n_1 and p_1 encode the trap energy level E_t through the Boltzmann relations

$$n_1 = N_C\, e^{-(E_C - E_t)/k_B T}, \qquad p_1 = N_V\, e^{-(E_t - E_V)/k_B T}, \qquad n_1 p_1 = n_i^2.$$

For approximately symmetric capture and at the injection levels typical of an operating cell, deep levels near midgap produce the strongest SRH recombination, because n_1 and p_1 are then smallest relative to n and p and the denominator is minimized. The statement is conditional in both respects: the rate-maximizing level depends on n, p, τ_n and τ_p, and shifts away from midgap for strongly asymmetric capture; and $n_1, p_1 \ll n, p$ is not guaranteed at midgap, since inside the depletion region or under low injection the carrier densities can fall to the order of n_i themselves. The asymmetric lifetimes $\tau_n \neq \tau_p$ model carrier-selective capture. In SolarLab the SRH parameters are *effective* quantities per layer unless trap-energy provenance is supplied.

7.5.2 Radiative Recombination And Photon Recycling

Band-to-band radiative recombination is bimolecular:

$$R_{\text{rad}} = B_{\text{rad}}(np - n_i^2). \tag{7.18}$$

In a high-refractive-index absorber most internally emitted photons are trapped by total internal reflection and reabsorbed, regenerating carriers — photon recycling. Following Yablonovitch's statistical ray optics, the single-pass escape probability of an internally emitted photon is

$$P_{\text{esc}} = \min\left(1,\ \frac{1}{4\, n_r^2\, \alpha(\lambda_g)\, d}\right), \tag{7.19}$$

evaluated at the band-edge wavelength $\lambda_g = hc/E_g$ from the absorber's tabulated n, k data, with d the absorber thickness. Only the escaping fraction is a true loss. SolarLab implements this at two levels of self-consistency:

1. **Net-coefficient form** (photon recycling on): the absorber's B_{rad} is rescaled by P_{esc} once at build time. Exact whenever the np product is spatially uniform across the absorber (the open-circuit regime where the V_{oc} boost is measured).
2. **Self-consistent reabsorption** (full tier): B_{rad} keeps its intrinsic value as a sink, and at every solver evaluation the total *net* emission $\int B_{\text{rad}}\,(np - n_i^2)\,dx$ over the absorber is redistributed as a uniform generation source $G_{\text{rad}} = (1 - P_{\text{esc}})\,\big[\int B_{\text{rad}}\,(np - n_i^2)\,dx\big]/d$. The net-emission integrand makes the recycled source vanish at mass-action equilibrium together with the sink, so the channel preserves detailed balance. This closes the recycling loop and retains the *global* coupling when np is non-uniform (strong injection, near-contact gradients, tandem sub-cells); it reduces to form 1 in the uniform limit. It is, however, a *spatially averaged* approximation: the redistribution is uniform over the absorber, with no spatial or spectral reabsorption kernel (emission position, propagation direction, wavelength-dependent absorption depth, and parasitic reabsorption are not resolved).

On the canonical radiative-limit configuration the recycling boost falls within the 40–100 mV regression acceptance window adopted for that configuration (motivated by reported MAPbI3 recycling effects, e.g. Pazos-Outón et al.). The recycling gain depends on the internal radiative efficiency, parasitic absorption, and out-coupling, so the window is a regression gate for the canonical preset rather than a general MAPbI3 figure.

7.5.3 Auger Recombination

$$R_{\text{Auger}} = (C_n n + C_p p)(np - n_i^2), \tag{7.20}$$

the three-particle channel that dominates at high carrier density (heavily doped layers, concentrator conditions, band-offset accumulation spikes).

The total bulk rate is:

$$R = R_{\text{SRH}} + R_{\text{rad}} + R_{\text{Auger}}. \tag{7.21}$$

7.5.4 Interface Recombination

Each internal electrical interface carries an areal SRH channel parameterized by surface recombination velocities (v_n, v_p):

$$R_s = \frac{n_s\, p_s - n_{i,\text{eff}}^2}{\dfrac{n_s + n_1}{v_p} + \dfrac{p_s + p_1}{v_n}} \qquad [m^{-2}s^{-1}], \tag{7.22}$$

converted to a volumetric sink at the interface node by dividing by the local dual-cell width. Three details matter for heterojunctions:

- **Cross-carrier sampling.** At a defect-bearing heterointerface the electron density is sampled on the transport-layer side and the hole density on the absorber side of the junction — the carrier populations that actually accumulate under cliff-type band offsets, following the polycrystalline-heterojunction treatment of Pauwels and Vanhoutte. This reproduces the characteristic cliff/spike asymmetry of interface-limited V_{oc}.
- **Detailed balance.** The reference $n_{i,\text{eff}}^2$ is the *product of the adjacent layers' equilibrium densities* ($n_{\text{eq},R}\, p_{\text{eq},L}$), not $n_{i,L}\, n_{i,R}$: with a heavily doped transport layer the equilibrium cross product exceeds the intrinsic product by many orders of magnitude, and a naive intrinsic reference would make the interface a spurious generator at dark equilibrium.
- **Non-negativity clamp (model-specific safeguard, defect-scoped).** At interfaces carrying a *declared defect* — where the cross-carrier approximation installs a bulk-asymptotic reference $n_{i,\text{eff}}^2$ that can exceed the true interface-plane product by orders of magnitude — the implementation clamps $R_s \geq 0$ to suppress the resulting spurious generation (of order 10 A/m^2 at illuminated short circuit). This clamp is *not* a general physical property: a real interface defect does generate carriers when $n_s p_s < n_{i,\text{eff}}^2$, the origin of depletion-region generation current and reverse dark current. Interfaces *without* a declared defect use the local mass-action reference and are left unclamped, so physical depletion generation is retained there. The clamp suppresses reverse-bias generation at declared-defect interfaces; an occupancy-consistent interface-state treatment would remove this restriction.

Optional refinements (default off, used for SCAPS cross-validation) evaluate the rate on Boltzmann-projected *interface-plane* densities, or solve a local supply-limited flux balance for the true plane densities, in which the plane carries the reduced interface gap $E_{g,s} = \min(E_C) - \max(E_V)$ across the junction — the mechanism behind the interface-gap rule of thumb $V_{\text{oc}} \lesssim (E_g - |\Delta E_C|)/q$.

A related optional correction (`het_recomb_despike`, a fraction $f \in [0, 1]$) addresses a discretization artifact: an abrupt valence-band offset produces a Boltzmann accumulation *spike* of majority carriers on the single interface node, and because $R_{\text{Auger}} \propto p^2 n$, the bulk Auger channel over-counts an interface loss that the areal channel of Eq. 7.22 already represents. The correction blends the junction-node densities toward the geometric mean of their neighbours *for the bulk-recombination evaluation only* — transport fluxes are untouched.

Assumptions and limits:

- SRH parameters are effective trap parameters unless trap-energy provenance is supplied;
- radiative and Auger coefficients should be layer-appropriate;
- interface recombination is localized numerically near the interface, so grid resolution matters when very large surface velocities are used.

7.6 Optical Generation

SolarLab supports two optical models of increasing fidelity.

7.6.1 Beer-Lambert Absorption

The scalar model uses a single absorption coefficient per layer:

$$G(x) = \Phi \alpha(x) \exp\left[-\int_0^x \alpha(x')\, dx'\right]. \tag{7.23}$$

It captures exponential attenuation of a monochromatic-equivalent photon flux Φ but no interference, reflection, or spectral structure.

7.6.2 Transfer-Matrix Optics

For wavelength-resolved generation, SolarLab implements the coherent transfer-matrix method (TMM) for multilayer thin films at normal incidence, following Pettersson et al. and Burkhard et al. Each layer is described by its complex refractive index $\tilde{n}(\lambda) = n + ik$ from tabulated n, k data. For every wavelength, 2×2 *interface* matrices built from the Fresnel coefficients

$$r_{ab} = \frac{\tilde{n}_a - \tilde{n}_b}{\tilde{n}_a + \tilde{n}_b}, \qquad t_{ab} = \frac{2\tilde{n}_a}{\tilde{n}_a + \tilde{n}_b},$$

alternate with *propagation* matrices carrying the complex phase $\delta_j = 2\pi \tilde{n}_j d_j / \lambda$ through each layer of thickness d_j. The assembled system matrix yields the stack reflectance and the forward/backward field amplitudes at every depth, from which the normalized internal intensity $|E(x, \lambda)|^2 / |E_0|^2$ follows. The local spectral absorption rate is

$$A(x,\lambda) = \frac{4\pi\, n(\lambda)\, k(\lambda)}{\lambda\, n_{\text{amb}}} \frac{|E(x,\lambda)|^2}{|E_0|^2} \qquad [m^{-1}], \tag{7.24}$$

where the n/n_{amb} factor converts field energy density to the Poynting-vector (energy-flux) normalization; this is the normalization under which the computed reflectance, transmittance, and layer-resolved absorptance satisfy $R + T + A = 1$, which the regression suite checks explicitly. The generation profile is the spectral integral against the incident AM1.5G photon flux,

$$G(x) = \int A(x,\lambda)\, \Phi_{\text{AM1.5G}}(\lambda)\, d\lambda, \tag{7.25}$$

evaluated over 300–1000 nm by default, assuming unit internal quantum efficiency of photogeneration, and cached once before the transient solve.

A millimetre-scale glass substrate is far thicker than the coherence length of sunlight, so treating it coherently would produce unphysical interference fringes. The first layer of the stack may therefore be flagged *incoherent*: it contributes a phase-free power-form Fresnel reflection and Beer-Lambert bulk attenuation, and hands the transmitted intensity to the coherent sub-stack beneath it.

EQE and electroluminescence require this wavelength-resolved machinery (monochromatic TMM generation plus a short-circuit drift-diffusion solve per wavelength); Beer-Lambert-only stacks do not contain sufficient spectral information for those experiments.

Assumptions and limits:

- Beer-Lambert optics are fast and useful for trends, but cannot represent interference, reflection, or wavelength-dependent collection;
- the TMM assumes planar layers, normal incidence, and coherent propagation (except the optional incoherent first substrate);
- TMM quality is limited by the provenance and wavelength coverage of the supplied optical constants;
- synthetic or placeholder optical constants should be treated as workflow demonstrations, not material-specific evidence.

7.7 Field-Dependent Mobility

Two empirical high-field mobility models can modify the low-field mobilities, composed multiplicatively and evaluated per face from the Poisson-solved electric field at every solver evaluation. The Caughey-Thomas form imposes velocity saturation,

$$\mu_{\text{CT}}(E) = \mu_0 \left[1 + \left(\frac{\mu_0 |E|}{v_{\text{sat}}}\right)^{\beta}\right]^{-1/\beta}, \tag{7.26}$$

with $\beta = 2$ (the Canali form, typical for electrons) and $\beta = 1$ (the Thornber form, typical for holes) as conventional exponents. The Poole-Frenkel form describes field-assisted barrier lowering in disordered or organic transport layers,

$$\mu_{\text{PF}}(E) = \mu_0 \exp\left(\gamma_{\text{PF}}\sqrt{|E|}\right), \tag{7.27}$$

with $\gamma_{\text{PF}} \sim 3 \times 10^{-4}\,(V/m)^{-1/2}$ representative of spiro-OMeTAD. Both hooks are opt-in per layer ($v_{\text{sat}} = 0$ and $\gamma_{\text{PF}} = 0$ disable them) and, because they must be re-evaluated from the instantaneous field, they are active only in the `full` tier.

7.8 Temperature Dependence

When temperature scaling is enabled, the device temperature T enters the model self-consistently. The thermal voltage $V_T = k_B T/q$ rescales every Einstein relation; carrier mobilities follow the standard scattering-style power law $\mu(T) = \mu_{300}\,(T/300\,K)^{\gamma_\mu}$; the radiative coefficient follows $B_{\text{rad}}(T) = B_{300}\,(T/300\,K)^{\gamma_B}$; and ionic diffusivity is Arrhenius-activated through the per-layer activation energy `E_a_ion`.

The band gap follows a Varshni shift *referenced to 300 K*, so that the configured E_g retains its meaning as the room-temperature gap:

$$E_g(T) = E_g^{300} - \alpha\left[\frac{T^2}{T+\beta} - \frac{T_0^2}{T_0+\beta}\right], \qquad T_0 = 300\,K. \tag{7.28}$$

Conventional semiconductors (silicon: $\alpha \approx 4.73 \times 10^{-4}$ eV/K, $\beta \approx 636$ K) narrow on heating; halide perovskites show the opposite sign and widen with temperature, reproduced by a negative α.

The band-edge effective densities of states follow the standard parabolic-band law,

$$N_C(T) = N_C^{300}\left(\frac{T}{300\,K}\right)^{3/2}, \qquad N_V(T) = N_V^{300}\left(\frac{T}{300\,K}\right)^{3/2}, \tag{7.29}$$

and the intrinsic density is recomputed from the shifted gap and the scaled densities of states, $n_i(T) = \sqrt{N_C(T)N_V(T)}\,e^{-E_g(T)/2k_BT}$. Because the SRH reference densities n_1, p_1 are configured at 300 K, they are rescaled by $n_i(T)/n_i(300)$ so that the mass-action invariant $n_1 p_1 = n_i^2(T)$ is preserved at every temperature; this holds the trap level fixed relative to midgap, the same convention used for graded layers. Without that rescaling the SRH reference densities would stay pinned at their 300 K values while $n_i(T)$ moved, which biases the dark and depletion-regime rates that a $V_{\text{oc}}(T)$ sweep reads.

Two quantities are deliberately *not* temperature-scaled, and both sit on default-off, 300 K-calibrated interface paths: the thermal velocity of the interface-plane state and closure formulations (fixed at the SCAPS value 10^7 cm s^{-1} rather than $v_{\text{th}} \propto T^{1/2}$), and the density-of-states prefactor of the per-side interface-defect trap levels. Cross-simulator $V_{\text{oc}}(T)$ comparisons that enable those options should hold $T = 300$ K or supply the temperature dependence explicitly.

All temperature hooks are gated by the temperature-scaling flag; the `legacy` tier pins the model at 300 K regardless of the configured T.

7.9 Spatial Trap Profiles

Grain-boundary and interface-adjacent defects concentrate near the transport layers. A per-layer trap-density profile with exponentially decaying edge enhancement,

$$N_t(x) = N_t^{\text{bulk}} + \left(N_t^{\text{if}} - N_t^{\text{bulk}}\right)\left(e^{-x/L_d} + e^{-(d-x)/L_d}\right), \tag{7.30}$$

(or a Gaussian variant with faster tail decay for defect slabs of well-defined width) is mapped onto the local SRH lifetimes through the inverse-proportionality of the SRH lifetime to trap density,

$$\tau(x) = \tau_{\text{bulk}} \frac{N_t^{\text{bulk}}}{\max\left(N_t(x),\, N_t^{\text{bulk}}\right)},$$

where the denominator floor caps the lifetime at τ_{bulk}: an edge-*passivation* profile ($N_t^{\text{if}} < N_t^{\text{bulk}}$) is clamped to the bulk lifetime rather than extrapolated to lifetimes above τ_{bulk}, so the profile can only ever add recombination. The two edge kernels are summed rather than maximized so that overlapping tails blend smoothly, and the sum is divided by its analytic supremum so that the shape function is bounded by unity and N_t saturates *at* N_t^{if}. Without that normalization the raw sum reaches $N_t^{\text{if}} + (N_t^{\text{if}} - N_t^{\text{bulk}})e^{-d/L_d}$ at each face, and $\simeq 2N_t^{\text{if}} - N_t^{\text{bulk}}$ for $d \ll L_d$ — up to a factor-of-two overshoot of the configured interface density. A single-face profile (`trap_edge` set to `left` or `right`) already peaks at exactly N_t^{if} and is not renormalized.

7.10 Band-Gap Grading

Compositionally graded absorbers (e.g. Ga-graded CIGS) are modelled with a *simplified subset* of the SCAPS material-interpolation framework of Burgelman and Marlein: a graded layer is a mixture of two endpoint materials A (front face) and B (back face) with a composition profile $y(x)$ that may be linear, parabolic, or an exponential notch. SCAPS itself admits a wider family — uniform, linear, logarithmic, parabolic, power-law, effective-medium, tabulated and Beta laws, selectable *per material parameter*, with the effective densities of states and the absorption coefficient gradeable alongside E_g and χ. Only the common subset should be compared across the two tools. The graded gap includes an optional bowing term,

$$E_g(y) = (1-y)\, E_g^A + y\, E_g^B - b\, y\,(1-y), \tag{7.31}$$

while the electron affinity interpolates linearly (Vegard's rule). The statistical quantities are anchored to the front face,

$$n_i^2(x) = n_{i,\text{front}}^2 \exp\left[-\frac{E_g(x) - E_g^{\text{front}}}{V_T}\right],$$

which is the ratio form of $n_i^2 = N_C N_V e^{-E_g/k_B T}$ under the assumption that $N_C N_V$ is *constant across the graded layer* — SolarLab does not grade the effective densities of states, so within this model the prefactors cancel exactly and no per-node DOS data are required. The cancellation is a consequence of that modelling choice, not a general identity: it would not hold against a tool that grades N_C and N_V with composition. The SRH references n_1, p_1 are graded so that $n_1 p_1 = n_i^2(x)$ holds node by node. Independent grading of χ and E_g controls the conduction- and valence-band profiles separately, which is how back-surface fields are configured. The transform is applied once at build time; a graded layer automatically refines its mesh so that a steep notch is resolved over several cells. Documented limitation: the *optical* constants are not graded — a graded absorber's absorption edge does not shift spatially, so J_{sc} reflects the electrical effect of grading under uniform optics only.

Chapter 8

Boundary And Initial Conditions

8.1 Potential

The potential is Dirichlet at both contacts:

$$\phi(0) = \phi_{\text{left}}, \qquad \phi(L) = \phi_{\text{left}} + V_{\text{bi}} - V_{\text{app}}.$$

Forward bias reduces the built-in field. By default the boundary value uses the configured `V_bi` (the IonMonger convention, in which V_{bi} is a free parameter representing the degenerate-doping limit). With the optional *flat-band-inspired* contact convention, the boundary instead uses the derived flat-band work-function difference `compute_V_bi()`, so the contact potential is fixed by the band alignment rather than by an input parameter. In one dimension only the work-function *difference* is physical, and this difference reproduces SCAPS's flat-band built-in potential exactly on the reference configuration; the convention is nevertheless not a full reproduction of the SCAPS contact model, which recomputes per-contact work functions with temperature and can include charge from deep defects in the contact-adjacent layers — neither is modelled here.

8.2 Carrier Contacts

By default, carrier densities at contacts are ohmic pins derived from local charge neutrality and mass action:

$$np = n_i^2, \qquad n - p = N_D - N_A.$$

The implementation evaluates the numerically stable two-branch closed form,

$$n_{\text{eq}} = \eta + \sqrt{\eta^2 + n_i^2}, \quad p_{\text{eq}} = \frac{n_i^2}{n_{\text{eq}}} \qquad \left(\eta = \tfrac{1}{2}(N_D - N_A) \geq 0\right),$$

and the mirrored branch for p-type material, which avoids catastrophic cancellation at heavy doping. An ohmic (Dirichlet) pin represents an ideal contact with infinite surface exchange velocity: the contact acts as an inexhaustible carrier reservoir at the dark doping equilibrium.

In `full` mode, selected contacts can instead use a finite-kinetics Robin flux boundary condition:

$$J = \pm qS(u - u_{\mathrm{eq}}),$$

where u is n or p, u_{eq} its contact equilibrium value, and S a surface recombination/extraction velocity. $S = 0$ is a perfectly blocking (Neumann) contact, finite S lets the boundary density evolve with a relaxation time $\sim \Delta x/S$, and $S \to \infty$ recovers the ohmic limit. Selective contacts are modelled by choosing a large S for the extracted carrier and a small S for the blocked carrier. For Schottky-type contacts the equilibrium reference can be taken as the thermionic reservoir density at the barrier, $n_{\mathrm{eq}} = N_C\, e^{-\phi_B/V_T}$ (and the N_V analogue for holes), which makes the contact carrier supply work-function-limited rather than doping-limited.

8.3 Ion Boundaries

Both positive and negative mobile ions use zero-flux boundaries. This reflects the physical assumption that ions redistribute inside the device but do not leave through the contacts; it also makes the total ion content an exact conserved quantity of the simulation.

8.4 Initial States

The dark equilibrium state starts from charge neutrality and mass action. The illuminated state is obtained by integrating the transient equations under illumination at the starting voltage until the coupled electron/hole/ion system reaches its light-soaked quasi-steady state.

For J-V, impedance, and degradation, this light-soaked initial state matters because ion and carrier memory affect the measurement: the same voltage grid scanned from a different pre-conditioning state produces a different transient current. Within this model that mobile-ion memory *is* the hysteresis mechanism, and the resulting loop is physical rather than a numerical artifact — but only once the time step, mesh, and convergence checks confirm it, since an unconverged transient produces a spurious loop of its own. It should not be read as a claim about experiment: measured J-V hysteresis in perovskite devices can also carry interface trapping, reversible polarization, slow contact processes, ion-electron coupling, measurement capacitance, and pre-conditioning differences, none of which this model separates.

Chapter 9

Physics Tiers

SolarLab uses three fidelity modes:

Mode	Active physics	Typical use
`legacy`	disables upgraded hooks	historical benchmark reproduction
`fast`	build-once upgrades, no expensive per-RHS hooks	screening
`full`	every configured hook	high-fidelity single runs

The distinction between `fast` and `full` is architectural: *build-once* physics is folded into the cached material arrays before the time integrator starts, while *per-evaluation* physics must be recomputed from the evolving state inside every right-hand-side call (Chapter 10). The flag matrix is:

Physics hook	`legacy`	`fast`	`full`	Evaluation
Band offsets + thermionic emission	off	on	on	build-once
Transfer-matrix optics	off	on	on	build-once
Dual mobile-ion species	off	on	on	build-once
Spatial trap profiles	off	on	on	build-once
Temperature scaling	off	on	on	build-once
Photon recycling (net coefficient)	off	on	on	build-once
Self-consistent radiative reabsorption	off	off	on	per-evaluation
Field-dependent mobility	off	off	on	per-evaluation
Selective/Robin contacts	off	off	on	per-evaluation

`legacy` disables every upgraded hook and is regression-tested against pinned IonMonger benchmark configurations and metrics (a specific configuration, sweep protocol, and metric envelope — see Chapter 16 — not a blanket equation-level reproduction claim across IonMonger's full parameter space).

One documented exception crosses the tier matrix: the device-level `flat_band_contacts` option activates the finite-kinetics Robin contact path on all four carrier/side channels on *every* tier, because the flat-band contact convention is defined by finite surface-exchange kinetics. The `off/off/on` row for selective contacts describes the explicit user-supplied `S_*` fields only.

The tier is a ceiling, not a command to fabricate missing data — a hook activates only when the configuration supplies the parameters it needs. For example:

- TMM requires `optical_material`.
- Field-dependent mobility requires nonzero `v_sat_*` or `pf_gamma_*`.
- Selective contacts require finite `S_*` values.
- Radiative reabsorption requires TMM/photon-recycling support.

This "enable-if-configured" pattern makes the tier strings safe to set on any preset: flags whose prerequisites are absent stay silent instead of forcing physics the stack cannot support.

Chapter 10

Numerical Method

The solver discretizes the governing equations of Chapter 7 on the device grid and then advances the coupled state in time. The important point for users is that a J-V point is not a closed-form diode equation; it is the terminal current read from a relaxed drift-diffusion state at a specified applied voltage.

10.1 Spatial Grid

The 1D solver builds a multilayer grid over the electrical layers only; substrate layers are excluded from the electrical grid but retained for TMM optics. Within each layer of thickness L the $N+1$ nodes follow a hyperbolic-tangent stretching,

$$x(\xi) = \frac{L}{2}\left[1 + \frac{\tanh(a\,\xi)}{\tanh a}\right], \qquad \xi \in [-1, 1],\ a = 3, \tag{10.1}$$

which clusters nodes toward both layer boundaries. This resolves the space-charge regions and interface carrier gradients — where densities vary over Debye-length scales — without wasting nodes in the quasi-neutral interior. Layer grids are concatenated with duplicate interface points removed, and a compositionally graded layer is automatically refined by an integer multiplier so a steep band-gap notch spans several cells.

10.2 Method Of Lines And State Vector

The PDEs are discretized in space first, producing a coupled system of ordinary differential equations in time — the method-of-lines approach. The packed state vector contains the nodal values

$$\mathbf{y} = (n,\ p,\ P\,[, P^{-}]\,[, \mathbf{s}_{\text{if}}]),$$

with the optional negative-ion block in dual-ion mode and an optional trailing block of interface-plane carrier states used by the steady-state SCAPS-comparison machinery. The electrostatic potential is *not* a dynamical unknown: Eq. 7.1 is elliptic and is re-solved exactly, from the instantaneous charge configuration, inside every right-hand-side (RHS) evaluation. This eliminates any splitting error between electrostatics and transport at the cost of one linear solve per evaluation, which the prefactored Poisson path below makes essentially free.

Solver Pipeline: How Each Timestep Works

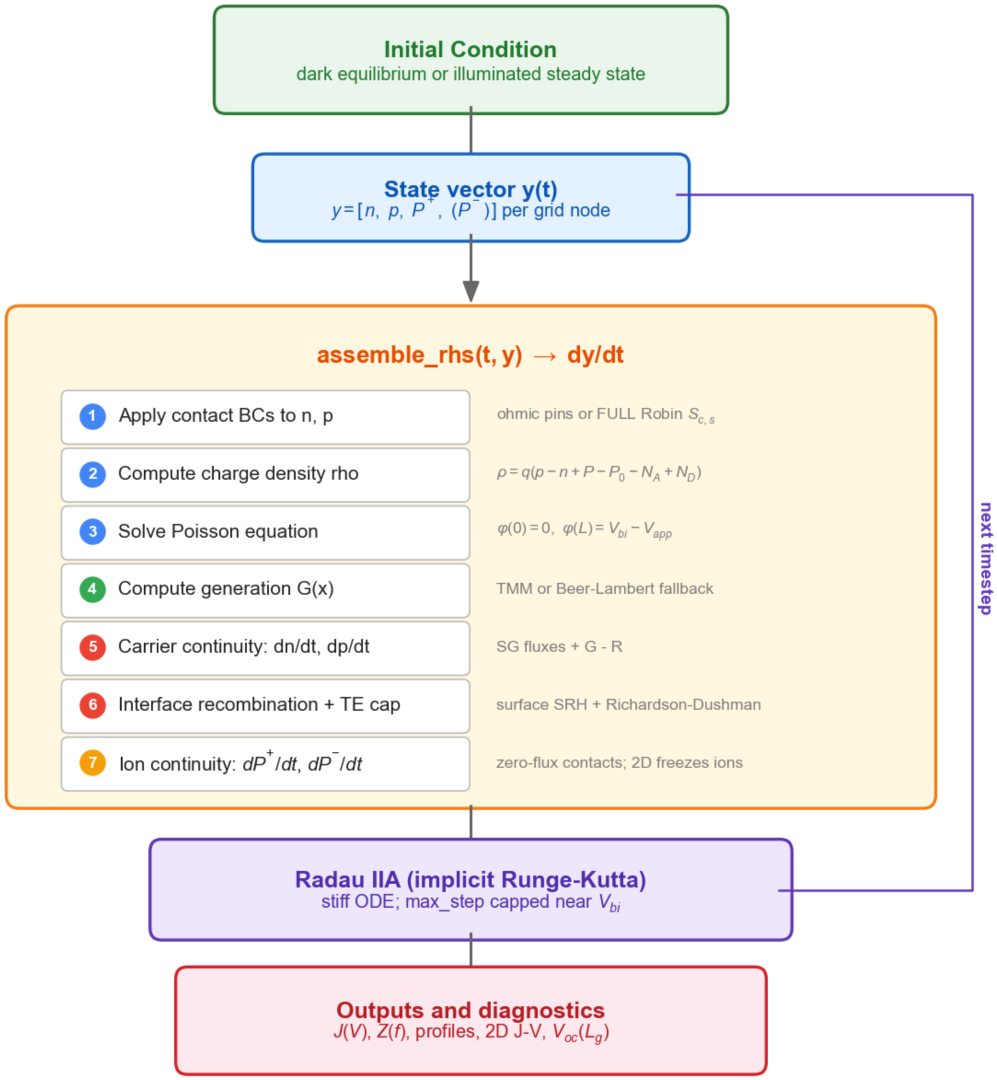

Figure 10.1: SolarLab solver pipeline

10.3 Scharfetter-Gummel Flux Discretization

Central differencing of the drift-diffusion current (Eq. 7.6) becomes unstable when the potential drop per cell exceeds a few V_T — the standard failure of naive discretizations in depletion regions. The Scharfetter-Gummel (SG) scheme instead solves the constitutive relation *exactly* on each cell under the assumptions of constant current and constant field between adjacent nodes; the resulting face flux is the optimal exponential fit,

$$J_{n,i+1/2} = \frac{qD_n}{\Delta x}\left[B(\xi)\, n_{i+1} - B(-\xi)\, n_i\right], \qquad \xi = \frac{\phi_{n,i+1} - \phi_{n,i}}{V_T}, \tag{10.2}$$

$$J_{p,i+1/2} = \frac{qD_p}{\Delta x}\left[B(\xi)\, p_i - B(-\xi)\, p_{i+1}\right], \qquad \xi = \frac{\phi_{p,i+1} - \phi_{p,i}}{V_T}, \tag{10.3}$$

with the Bernoulli function

$$B(\xi) = \frac{\xi}{e^{\xi} - 1}. \tag{10.4}$$

The scheme reduces to central differencing in the diffusion limit ($|\xi| \ll 1$) and to pure upwinding in the drift limit ($|\xi| \gg 1$), remaining stable at arbitrarily large cell Peclet numbers (Scharfetter and Gummel 1969; Selberherr 1984). Its positivity property is a statement about the *spatial semi-discretization* — the face flux is monotone, so the semi-discrete operator does not manufacture negative densities — and does not carry over automatically to the fully discrete transient: the state vector (n, p, P) is advanced by a general-purpose implicit integrator, which does not enforce non-negativity at each internal stage or accepted step, and SolarLab applies no clipping or projection to n and p during the transient. Positivity *is* enforced by construction in the direct steady-state driver, which solves in logarithmic unknowns (Chapter 10). The scheme rests on three implementation details:

- the drift argument ξ is built from the *band-effective* potentials ϕ_n, ϕ_p of Eqs. 7.8–7.9, so heterojunction offsets and density-of-states contrasts enter the flux without any special-case interface treatment;
- $B(\xi)$ is evaluated in three numerically safe branches: a Taylor series for $|\xi| < 10^{-8}$ (removing the $0/0$ singularity), the analytic zero limit for $\xi > 700$ (preventing floating-point overflow of e^{ξ}), and the `expm1` form elsewhere (preserving precision near zero);
- face diffusivities use the harmonic mean of the adjacent nodal values, the composition consistent with flux continuity across a material discontinuity (it also preserves exact zeros, so an ion-blocking layer boundary carries strictly zero ionic flux).

The divergence of the face fluxes is taken over dual-grid (finite-volume) cells, so the semi-discrete continuity equations conserve charge exactly on the non-uniform mesh. The same SG form, with the drift sign set by the ionic charge and the steric factor of Eq. 7.16 folded into the face diffusivity, discretizes the ion flux of Eq. 7.14.

10.4 Poisson Fast Path

The finite-volume Poisson operator is tridiagonal. Its LU factorization depends only on the grid and the permittivity profile — both constant for the lifetime of an experiment — so it is computed once (LAPACK `dgttrf`) and cached; every subsequent RHS evaluation performs a single $O(N)$ back-substitution (`dgttrs`) with the Dirichlet boundary values absorbed into the right-hand side. This is roughly forty times faster than a general sparse solve and is the reason the per-experiment material cache must be built *outside* the time-step loop.

The same build-once strategy applies to all static coefficient fields: per-node and per-face material arrays, band-edge and density-of-states folds, trap-profile lifetime maps, grading profiles, tunnelling-enhanced Richardson constants, and the TMM generation profile are all assembled once into an immutable cache. Only three hooks are inherently state-dependent and re-evaluated per RHS call — field-dependent mobility, self-consistent radiative reabsorption, and Robin contact fluxes — which is exactly the set excluded from the `fast` tier.

10.5 Right-Hand-Side Assembly

Each RHS evaluation proceeds in a fixed order:

1. apply contact conditions to the carrier arrays (ohmic Dirichlet pins, or free boundary densities on Robin sides);
2. assemble the space charge ρ (Eq. 7.2);
3. solve Poisson for ϕ via the prefactored path;
4. form the generation profile $G(x)$ (cached TMM or Beer-Lambert; plus the reabsorption source when self-consistent photon recycling is active);
5. recompute face diffusivities from the face field if field-dependent mobility is active;
6. evaluate SG fluxes, apply the thermionic-emission cap at flagged heterointerface faces, and take flux divergences with $G - R$;
7. subtract interface recombination at interface nodes;
8. evaluate ion continuity (and the negative species in dual-ion mode);
9. enforce the boundary treatment (zero time-derivative on Dirichlet-pinned entries; Robin fluxes otherwise), and verify all derivatives are finite.

10.6 Time Integration And Robustness

The semi-discrete system is stiff: electronic relaxation (nanoseconds), dielectric relaxation, and ionic migration (milliseconds-seconds) coexist in one state vector. SolarLab integrates it with SciPy's Radau IIA method — a fifth-order, L-stable, fully implicit Runge-Kutta scheme appropriate for stiff differential systems (Hairer and Wanner 1996) — with BDF as a fallback. Stability of the implicit iteration, not accuracy, sets the practical step size, and a layered set of safeguards keeps difficult J-V steps bounded:

- **Step-size ceiling near flat band.** Close to $V_{\text{app}} \approx V_{\text{bi}}$ the Jacobian is nearly singular and Radau's local-error estimator can under-report the truncation error, accepting a single oversized step onto a spurious branch of the implicit equations (visible as an isolated non-physical J-V spike). Every voltage step therefore caps the internal step at $1/20$ of the settle interval.
- **Evaluation budget.** A per-step ceiling on RHS evaluations converts a non-contracting implicit iteration (which would otherwise spin without wall-time bound) into a recoverable failure.
- **Bisection in time.** A failed settle interval is halved and re-chained, up to 2^{10} sub-intervals.
- **Frozen-source retry.** The self-consistent reabsorption source couples all absorber nodes through a non-local integral; at the diode-injection knee this can prevent Newton contraction inside Radau. The failed step is retried with the reabsorption source frozen at its entry-state value — a lag bounded by one settle interval, well below the regression tolerance.

- **Method fallback.** A step that exhausts bisection is attempted once with BDF, which occasionally converges where Radau's error estimator stalls.
- **Finite-value guard.** Any NaN/Inf in the assembled derivatives raises immediately rather than letting the integrator accept corrupted state.

10.7 Direct Steady-State Solver

The transient integrator is the reference engine for ion-coupled dynamics, but steady-state comparisons (e.g. against ion-free steady-state simulators such as SCAPS) and artifact-free open-circuit voltages benefit from solving $F(\mathbf{y}) = 0$ directly on the *same* discretized RHS — one physics implementation, two drivers. The steady-state driver freezes the ion profile from its seed state and applies a damped Newton method with:

- **Logarithmic unknowns** $z = (\ln n, \ln p)$, which enforce positivity by construction and equidistribute sensitivity across densities spanning twenty orders of magnitude;
- **Peak-density residual scaling**: residuals are normalized by the global peak density rather than the local one, because at dark depletion nodes the local-relative rate floors at double-precision cancellation noise and no tolerance could ever be met; on the peak-density scale that noise sits near 10^{-10} while a physical imbalance registers at order unity, and the tolerance translates into a terminal-current error bound of $\sim 10^{-8}\,A\,m^{-2}$;
- **Finite-difference Jacobian with chord reuse**: the dense Jacobian is built by forward differences and its LU factorization is reused across chord iterations, rebuilt only when a stale factor loses contraction — the dominant cost saving on warm-started continuation solves;
- **Backtracking line search** on the residual 2-norm, with the Newton step capped in log space;
- **Smoothed thermionic-emission cap**: the hard magnitude-minimum of the TE cap is non-differentiable exactly where it binds; the steady-state driver (and only it) replaces it with a narrow sigmoid blend so Newton retains a usable derivative;
- **Quasi-Fermi-preserving Poisson relaxation**: before Newton starts (and after every transient assist), a nonlinear Poisson solve with Boltzmann-responding carriers at frozen quasi-Fermi levels performs the dielectric relaxation analytically. This step is unconditionally well-posed and removes the seconds-slow relaxation mode that otherwise dominates the Jacobian conditioning in near-insulating layers;
- **Certified fallbacks**: where the TE-cap kink makes Newton stall at an already-physically-converged state, the best iterate is accepted under an explicit residual bound; where Newton cannot finish at all, escalating Radau settles compute the point and are *certified* against the same residual scale; an uncertifiable point raises an error rather than returning an unconverged result.

Steady-state J-V curves are computed by voltage continuation with warm-started seeds and automatic voltage-step bisection; the open-circuit voltage can additionally be located by direct bisection on the steady-state $J(V)$ zero crossing, eliminating sweep-grid interpolation error. Measured agreement between the two drivers on the reference configuration is within 5 mV in V_{oc} and 1% in J_{sc}. Two deep-tail regimes are handled by the transient engine rather than the Newton driver: near-insulating contact doping produces no $J = 0$ crossing below the scan ceiling (a physical absence the transient confirms), and deeply collapsed conduction-band-offset junctions are pathological for any algebraic Newton method and are handled by the transient engine only.

10.8 Operator Splitting For Slow Ion Dynamics

Long-time experiments (degradation) use an operator-splitting step matched to the two-time-scale structure: the ion subsystem is advanced over a macroscopic interval with the carrier densities frozen (Poisson re-solved inside the ion RHS, so the ions always see a field consistent with the *current ion profile and the frozen carriers*), and the carriers are then re-equilibrated by a short full-system transient of order one carrier lifetime. Re-solving Poisson makes the potential consistent with the instantaneous charge density; it does not keep the electrons and holes quasi-stationary with respect to the evolving ionic field, so this is a first-order operator splitting rather than a fully self-consistent electron-ion advance. Its error grows with the macro-step and with the band-bending change the ions produce over that step, so a macro-step halving study against the fully coupled solution is the appropriate check before quoting long-time results — most importantly near open circuit and near flat band, where the ionic screening moves the bands the furthest. Snapshot J-V measurements inside degradation freeze the ion profile entirely, so each snapshot reports the instantaneous electronic response of the current ionic configuration.

10.9 Terminal Current Evaluation

The reported current density combines conduction, ionic, and displacement contributions at mesh faces:

$$J = J_n + J_p + J_{\text{ion}} + \varepsilon_0 \varepsilon_r \frac{\partial E}{\partial t}.$$

Transient sweeps read the contact-adjacent face, where the displacement term is part of the physical external-circuit current. Steady-state probes (Suns-V_{oc}, EQE) instead take the *median* across interior faces: charge conservation makes $J(x)$ uniform at steady state, so any face is formally correct, but residual ionic and displacement transients concentrate at the boundary faces and the median is robust to those outliers.

Because a median summarises the face-to-face disagreement rather than removing it, both steady-state probes also report how large that disagreement was,

$$\Delta J_{\text{spread}} = \max_x J(x) - \min_x J(x),$$

evaluated over the interior faces only — the two contact-adjacent faces are excluded because their displacement term is physical external-circuit current and legitimately swings on ionic-rich stacks, so including them would report a real boundary transient as a conservation error. The value is returned as `J_spread_max` on both result objects.

It is deliberately reported rather than enforced. Charge conservation does make $J(x)$ uniform at an exact steady state, which suggests using ΔJ_{spread} as an acceptance threshold for V_{oc}, J_{sc} and FF — but measured on the TMM presets at $V = 0$ that threshold is not defensible. The reported median moves by less than 0.02 % between a 1 ms and a 100 ms settle, while over the same change the spread on the IonMonger TMM preset *grows* by a factor of 43, and it swings by roughly 80x between 12 and 20 nodes per layer at fixed settling time. Normalising by the largest individual current component does not repair the behaviour. The cause is the terminal-current flux cancellation discussed above: J is a sum of large, nearly-cancelling electron, hole, ionic and displacement contributions, so a small relative error in the parts appears as a large absolute error in their sum. That is precisely the regime in which the median is the appropriate summary and the raw spread is a poor convergence proxy — a fixed tolerance would reject converged runs, and its natural remedy of settling longer makes the number worse. Use ΔJ_{spread} as a magnitude diagnostic, and establish convergence by refining the mesh and confirming that the *extracted metrics* stop moving. Metrics extraction interpolates J_{sc} at $V = 0$ and V_{oc} at the first positive-to-negative zero crossing, computes FF and PCE from the maximum-power point over the operating quadrant, and reports an explicit `voc_bracketed` flag rather than metrics extrapolated from a sweep that does not bracket the crossing. The hysteresis index compares scan directions as

$$\text{HI} = \frac{PCE_{\text{rev}} - PCE_{\text{fwd}}}{PCE_{\text{rev}}}.$$

10.10 Two-Dimensional Extension

The experimental 2D solver reuses the same physics on a tensor-product grid: the tanh-clustered stack axis crossed with a uniform lateral axis. The 2D Poisson operator is a five-point finite-volume stencil with harmonic-mean face permittivities, factorized once with a sparse LU and reused per evaluation; the carrier fluxes are vectorized 2D SG forms with the same band-effective potentials, and the thermionic-emission cap applies on vertical (stack-direction) heterointerface faces. Mobile ions are held as a static Poisson background extruded from the 1D solution — the documented 2D model assumption. Lateral boundaries are periodic or zero-flux. The 2D solver is regression-gated against 1D in the lateral-uniform limit to $|\Delta V_{\text{oc}}| \leq 0.1$ mV, relative $\Delta J_{\text{sc}} \leq 5 \times 10^{-4}$, and $|\Delta FF| \leq 10^{-3}$.

10.11 Numerical Safety

Important safety mechanisms include:

- finite-value guards in the RHS;
- harmonic face permittivity;
- stable Bernoulli-function branches;
- the thermionic-emission cap at abrupt heterointerfaces;
- step-size ceilings, evaluation budgets, and bisection/fallback logic in every voltage-stepping experiment;
- TMM determinant guard;
- median-current steady-state readout;
- explicit `voc_bracketed` flags when a J-V sweep misses the zero crossing;
- steady-state convergence raises on non-convergence.

For publication work, numerical settings are part of the method. Report the grid size, voltage spacing, settling time or scan rate, tolerances when modified, and whether the run used `legacy`, `fast`, or `full` mode.

Chapter 11

Running SolarLab

11.1 Installation

From `perovskite-sim/`:

```
pip install -e ".[dev]"
```

For the frontend:

```
cd frontend
npm install
```

11.2 Backend

From the SolarLab root:

```
uvicorn backend.main:app \
  --host 127.0.0.1 --port 8000 \
  --app-dir perovskite-sim --reload
```

Do not run `uvicorn main:app` from inside `backend/`; the backend imports are written around `--app-dir perovskite-sim`.

11.3 Frontend

```
cd perovskite-sim/frontend
npm run dev
```

Then open:

```
http://127.0.0.1:5173
```

11.4 Python API

```
from perovskite_sim.models.config_loader import load_device_from_yaml
from perovskite_sim.experiments.jv_sweep import run_jv_sweep
```

```
stack = load_device_from_yaml("configs/nip_MAPbI3.yaml")
result = run_jv_sweep(stack, N_grid=80, n_points=40, v_rate=1.0)

print(result.metrics_fwd.V_oc)
print(result.metrics_fwd.J_sc)
print(result.metrics_fwd.FF)
print(result.metrics_fwd.PCE)
```

11.5 Worked Example 1: Baseline J-V Sweep

This example runs a standard n-i-p MAPbI3 J-V sweep and reports the principal photovoltaic metrics.

```
cd perovskite-sim
python - <<'PY'
from perovskite_sim.models.config_loader import load_device_from_yaml
from perovskite_sim.experiments.jv_sweep import run_jv_sweep

stack = load_device_from_yaml("configs/nip_MAPbI3.yaml")
result = run_jv_sweep(stack, N_grid=80, n_points=40, v_rate=1.0)
m = result.metrics_fwd
print(f"Voc = {m.V_oc:.4f} V")
print(f"Jsc = {m.J_sc:.2f} A/m^2")
print(f"FF  = {m.FF:.3f}")
print(f"PCE = {100*m.PCE:.2f} %")
print(f"Voc bracketed: {m.voc_bracketed}")
PY
```

Metric interpretation:

- J_{sc} is the current density near zero applied voltage.
- V_{oc} is interpolated where the terminal current crosses zero.
- FF and PCE are meaningful only when `voc_bracketed` is true.
- Forward and reverse metrics can differ when mobile ions retain scan history.

11.6 Worked Example 2: Absorber Thickness Study

This example varies absorber thickness while leaving the rest of the stack unchanged, allowing optical collection and recombination trends to be compared under controlled assumptions.

```
from dataclasses import replace
from perovskite_sim.models.config_loader import load_device_from_yaml
from perovskite_sim.experiments.jv_sweep import run_jv_sweep

base = load_device_from_yaml("configs/nip_MAPbI3.yaml")
absorber_index = next(i for i, layer in enumerate(base.layers)
                      if layer.role == "absorber")

for thickness_nm in (200, 400, 700, 1000):
    layers = list(base.layers)
    layers[absorber_index] = replace(
```

```
        layers[absorber_index],
        thickness=thickness_nm * 1e-9,
    )
    stack = replace(base, layers=tuple(layers))
    result = run_jv_sweep(stack, N_grid=80, n_points=30, v_rate=2.0)
    m = result.metrics_fwd
    print(thickness_nm, m.V_oc, m.J_sc, m.FF, m.PCE)
```

Expected interpretation:

- Thicker absorbers can absorb more light, but can also increase recombination or transport loss.
- If J_{sc} does not change under Beer-Lambert optics, check whether `alpha` and `Phi` are physically configured.
- If FF collapses, inspect spatial profiles and contact selectivity before concluding that the absorber is intrinsically poor.

11.7 Worked Example 3: Enabling TMM Optics

This example compares scalar Beer-Lambert generation with wavelength-resolved transfer-matrix generation.

```
from perovskite_sim.models.config_loader import load_device_from_yaml
from perovskite_sim.experiments.jv_sweep import run_jv_sweep

for config in ("configs/nip_MAPbI3.yaml", "configs/nip_MAPbI3_tmm.yaml"):
    stack = load_device_from_yaml(config)
    result = run_jv_sweep(stack, N_grid=60, n_points=21)
    print(config, result.metrics_fwd.J_sc)
```

Expected interpretation:

- TMM includes interference, reflection, parasitic absorption, and spectral weighting.
- TMM results are only as credible as the `optical_material` data and AM1.5G spectrum used.
- EQE and EL should use TMM-enabled configurations, not Beer-Lambert-only presets.

Chapter 12

Backend API

Configuration endpoints:

Method	Path	Purpose
`GET`	`/api/configs`	list shipped and user presets
`GET`	`/api/configs/{name}`	load one preset
`POST`	`/api/configs/user`	save a user-edited stack
`GET`	`/api/layer-templates`	layer template library
`GET`	`/api/optical-materials`	available TMM optical keys

Preferred experiment interface:

Method	Path	Purpose
`POST`	`/api/jobs`	submit `{kind, config_path|device, params}`
`GET`	`/api/jobs/{id}/events`	stream progress/result/error/done events

Supported job kinds:

```
jv, current_decomp, spatial, dark_jv, suns_voc, voc_t,
eqe, el, impedance, tpv, degradation, tandem,
mott_schottky, jv_2d, voc_grain_sweep
```

The frontend primarily uses streaming jobs. Long-running experiments should report progress through callbacks that the backend converts to SSE frames.

Chapter 13

Experiment Manual

13.1 J-V Sweep

The J-V sweep runs forward and reverse voltage scans while preserving state history. This state-preserving scan is the implemented representation of hysteresis in an ion-coupled perovskite cell.

Outputs:

- V_fwd, J_fwd;
- V_rev, J_rev;
- metrics_fwd, metrics_rev;
- hysteresis_index.

Metrics:

$$FF = \frac{P_{\text{mpp}}}{V_{\text{oc}} J_{\text{sc}}}.$$

$$PCE = \frac{P_{\text{mpp}}}{P_{\text{in}}}, \qquad P_{\text{in}} = 1000\, W\, m^{-2} \text{ by default (AM1.5G, 1 sun).}$$

The metric extractor accepts an explicit P_in; runs that change the illumination intensity or spectrum (e.g. through Phi or intensity sweeps) must supply the actual incident power density, or the reported efficiency refers to the 1-sun denominator rather than the simulated illumination.

If the sweep never crosses $J = 0$, voc_bracketed=false. In that case V_{oc}, FF, and PCE are sentinel values and the user should increase V_max.

The J-V job additionally accepts a solver selector: the default transient driver is the state-preserving Radau sweep described above, while steady_state routes the same device through the ion-frozen direct Newton driver of Chapter 10, returning a single hysteresis-free steady-state curve. The steady-state driver is the appropriate choice when comparing against ion-free steady-state simulators.

13.2 Current Decomposition

Current decomposition separates:

- electron current J_n;
- hole current J_p;
- ion current J_{ion};
- displacement current J_{disp};
- total current.

This separation helps identify whether a transient is dominated by electronic, ionic, or capacitive response.

13.3 Spatial Profiles

Spatial-profile runs save snapshots of:

- x;
- ϕ;
- E;
- n;
- p;
- P;
- ρ;
- applied voltage.

These profiles are essential for interpreting unusual J-V behavior. For example, a high V_{oc} with poor FF may indicate selective-contact or recombination issues rather than poor optical generation.

13.4 Dark J-V

Dark J-V fits the diode equation:

$$J = J_0 \left[\text{exp} \left(\frac{V}{n_{\text{id}} V_T} \right) - 1 \right].$$

The output includes ideality factor n_{id}, saturation current J_0, and the fitted voltage window.

13.5 Suns-V_{oc}

Suns-V_{oc} varies illumination intensity and extracts V_{oc} at each level. It constructs a pseudo-JV curve:

$$J_{\text{pseudo}}(X) = J_{\text{sc,ref}} - J_{\text{sc}}(X), \qquad V_{\text{pseudo}}(X) = V_{\text{oc}}(X).$$

The pseudo-JV representation is less affected by series resistance than a standard terminal J-V sweep.

13.6 $V_{\text{oc}}(T)$

The temperature sweep estimates recombination activation energy by fitting V_{oc} versus temperature:

$$V_{\text{oc}}(T) \approx \frac{E_A}{q} - \frac{AkT}{q} \ln\left(\frac{J_{00}}{J_{\text{sc}}}\right),$$

which follows from the single-diode forms $J = J_0 \exp(qV/Ak_BT)$ and $J_0 = J_{00} \exp(-E_A/Ak_BT)$ with ideality factor A.

The intercept at $T = 0$ is reported as E_A in eV. Note that a *temperature-independent* A enters the slope only and leaves that intercept unchanged, so the extracted activation energy is insensitive to the value of A — it is $A(T)$, $J_{\text{sc}}(T)$ and $E_g(T)$ varying across the fitted window that bias it. The linear extrapolation to 0 K is therefore an empirical activation energy for the fitted temperature range, not a zero-temperature physical quantity; report the range, the fit quality, and whether J_{sc} was allowed to vary alongside it.

When comparing $V_{\text{oc}}(T)$ against other simulators, the temperature models are generally not equivalent: SolarLab applies $E_g(T)$, $\mu(T)$, $B_{\text{rad}}(T)$, $n_i(T)$, and Arrhenius ion scaling automatically when temperature scaling is enabled, whereas reference tools (e.g. SCAPS) treat only a subset of quantities as temperature-dependent by default and expect the rest to be supplied per temperature. Align every temperature-dependent parameter explicitly before drawing cross-simulator conclusions.

13.7 EQE / IPCE

EQE is:

$$\text{EQE}(\lambda) = \frac{J(\lambda) - J_{\text{dark}}}{q\Phi_{\text{inc}}(\lambda)}.$$

The incremental current is used *signed* against the solar convention ($J > 0$ at $V = 0$ under illumination), so a negative EQE is reported as such and raises a runtime warning rather than being hidden by an absolute value: a wrong-way incremental current indicates a contact polarity, sign-convention, or dark-baseline convergence problem, and must not be presented as a physical quantum efficiency.

SolarLab computes monochromatic TMM generation, solves the drift-diffusion problem at short circuit, subtracts a dark baseline current J_{dark} (the device settled at $V = 0$ with $G = 0$, which cancels any residual ionic drift that has not fully damped), and integrates against AM1.5G for J_{sc}. The incremental ΔJ definition matches the standard measurement convention. The baseline state is the settled dark short-circuit state; bias-light protocols are not modelled. This experiment requires `optical_material`.

13.8 Electroluminescence

EL uses the Rau reciprocity relation, whose general form is

$$\Phi_{\text{EL}}(\lambda, V) = \text{EQE}_{\text{PV}}(\lambda)\, \phi_{\text{bb}}(\lambda, T) \left[\exp\left(\frac{qV}{kT}\right) - 1\right],$$

where EQE_{PV} is the external photovoltaic quantum efficiency (absorption *and* collection). SolarLab evaluates the absorber-absorptance upper bound of this relation,

$$\Phi_{\mathrm{EL}}(\lambda) \approx A_{\mathrm{abs}}(\lambda)\phi_{\mathrm{bb}}(\lambda, T)\,\mathrm{exp}\left(\frac{qV_{\mathrm{inj}}}{kT}\right),$$

which assumes near-unity carrier collection ($\mathrm{EQE}_{\mathrm{PV}} \approx A_{\mathrm{abs}}$), a flat quasi-Fermi-level splitting equal to qV_{inj} across the absorber, and injection well above thermal ($e^{qV/kT} \gg 1$, so the -1 is dropped). Reciprocity itself further assumes transport that is linear in the excess carrier density and only weakly dependent on the operating point.

The blackbody photon flux is:

$$\phi_{\mathrm{bb}}(\lambda, T) = \frac{2\pi c/\lambda^4}{\mathrm{exp}(hc/\lambda kT) - 1}.$$

The non-radiative voltage loss is:

$$\Delta V_{\mathrm{nr}} = -V_T \ln(\mathrm{EQE}_{\mathrm{EL}}),$$

where $\mathrm{EQE}_{\mathrm{EL}}$ is the external electroluminescence quantum efficiency (emitted radiative flux over injected carrier flux) evaluated at the injection point corresponding to open circuit. The relation quantifies the V_{oc} deficit relative to the radiative limit under the reciprocity assumptions above; it is not an unconditional identity at arbitrary injection. This experiment also requires TMM optical data.

13.9 Impedance

Impedance applies a small sinusoidal voltage perturbation around a DC bias, integrates a few AC cycles of the full nonlinear transient at each frequency, and extracts the amplitude and phase of the current response by lock-in demodulation (sine/cosine multiplication followed by low-pass filtering). The displacement current $\varepsilon_0\varepsilon_r\,\partial E/\partial t$ is included, so the capacitive branch is physical rather than post-processed. The result is $Z(f)$ in $\Omega\,m^2$, suitable for Nyquist and Bode plots.

This is a *time-domain, small-amplitude, full-nonlinear-model* method — it is not a linearized small-signal AC solver of the kind SCAPS implements, and results from the two approaches are comparable only in the linear small-signal regime. Because the ionic and electronic subsystems respond on different time scales, the spectrum can resolve both relaxations without an equivalent-circuit assumption, provided the selected frequency window spans both time constants and the per-frequency integration reaches a periodic steady state; too few settling cycles or an excessive perturbation amplitude degrade the extracted spectrum.

13.10 Mott-Schottky

Mott-Schottky wraps dark impedance at a fixed frequency and fits:

$$\frac{1}{C^2} = aV + b.$$

Then:

$$V_{\text{bi}}^{\text{app}} = -\frac{b}{a} + \frac{k_B T}{q}, \qquad N_{\text{eff}}^{\text{app}} = -\frac{2}{q\varepsilon_r\varepsilon_0 a}.$$

The $k_B T/q$ term is the majority-carrier tail at the depletion edge: the one-sided abrupt-junction capacitance is $1/C^2 = 2(V_{\text{bi}} - V - k_B T/q)/(q\varepsilon_s\varepsilon_0 N)$, so the bare V-axis intercept $-b/a$ corresponds to $V_{\text{bi}} - k_B T/q$, i.e. 25.9 mV low at 300 K. A depletion capacitance has $a < 0$, which makes $N_{\text{eff}}^{\text{app}}$ positive; the fit does not determine the doping *type*, which must be read off the stack and reported separately from $|N|$.

Both extracted quantities are *apparent* values: the textbook interpretation assumes a one-sided abrupt junction, uniform doping, a known permittivity, a cleanly separable depletion capacitance, and traps/ions that do not respond at the measurement frequency. In thin-film perovskites the geometric capacitance, injection capacitance, interface states, and ionic response can each produce a near-linear $1/C^2$ slope of their own, so for thin fully depleted absorbers the analysis can over-interpret non-depletion response. Report the fit frequency, bias window, and phase alongside the fitted values, and do not report the slopes automatically as physical dopant densities.

13.11 Transient Photovoltage

TPV applies a small generation pulse at open circuit. SolarLab enforces the open-circuit condition by adjusting V_{app} so that terminal current remains near zero at each reported time. The voltage decay is fitted as:

$$V(t) \approx V_{\text{oc}} + \Delta V_0 \exp(-t/\tau).$$

13.12 Degradation

The degradation experiment is a long-time ion-coupled damage proxy. It advances the ion/carrier system under bias and periodically performs snapshot J-V measurements. During each snapshot, ions are frozen while carriers relax, separating slow ionic redistribution from the instantaneous electronic response.

13.13 Tandem J-V

The tandem driver:

1. runs one combined TMM optical calculation over the full tandem stack;
2. runs independent top and bottom sub-cell J-V sweeps with fixed generation;
3. series-matches at a common current;
4. sums the sub-cell voltages.

This is a SolarLab-native workflow (reference steady-state simulators such as SCAPS handle multi-junction devices by external scripting rather than a built-in driver, so no cross-simulator parity is implied). The current driver neglects luminescent coupling between sub-cells, tunnel-junction resistance, and any electrical inter-cell coupling beyond ideal series current matching.

Some tandem optical constants are documented as placeholder/stub data. These must be replaced before using the tandem driver for material-specific predictions.

13.14 2D J-V

The 2D J-V driver uses a tensor-product grid. It is validated against 1D in the lateral-uniform limit and can add absorber grain boundaries with reduced SRH lifetimes. Ions are frozen as static Poisson background in 2D.

13.15 $V_{\text{oc}}(L_g)$ Grain Sweep

The grain sweep repeats 2D J-V simulations over grain sizes L_g and reports V_{oc}, J_{sc}, and FF as functions of grain size. It is intended to quantify microstructure-driven voltage loss within the frozen-ion 2D approximation.

Chapter 14

Shipped Presets

Representative shipped presets:

Preset	Purpose
`nip_MAPbI3.yaml`	n-i-p MAPbI3, Beer-Lambert optics
`nip_MAPbI3_tmm.yaml`	n-i-p MAPbI3 with TMM optics
`pin_MAPbI3.yaml`	p-i-n MAPbI3, Beer-Lambert optics
`pin_MAPbI3_tmm.yaml`	p-i-n MAPbI3 with TMM optics
`ionmonger_benchmark.yaml`	Courtier/IonMonger-style benchmark
`driftfusion_benchmark.yaml`	Driftfusion-inspired benchmark
`cigs_baseline.yaml`	CIGS-like inorganic thin-film stack
`cSi_homojunction.yaml`	crystalline silicon homojunction
`radiative_limit.yaml`	photon-recycling and radiative-limit checks
`selective_contacts_demo.yaml`	Robin/selective-contact demonstration
`field_mobility_demo.yaml`	field-dependent mobility demonstration
`tandem_lin2019.yaml`	2T tandem workflow demonstration
`twod/nip_MAPbI3_uniform.yaml`	2D lateral-uniform parity preset
`twod/nip_MAPbI3_singleGB.yaml`	2D single grain-boundary preset
`twod/bcx_combined_demo.yaml`	combined 2D advanced-physics demo

Preset values are simulation inputs, not universal material constants. For publication studies, users should document the source of every parameter they change or use.

Chapter 15

Troubleshooting And Diagnostics

15.1 `voc_bracketed=false`

Meaning: the J-V sweep did not find a current zero crossing inside the sampled voltage range.

Likely causes:

- `V_max` is too low;
- the voltage grid is too coarse near open circuit;
- the device is numerically unstable near flat band;
- contact or recombination settings produce an unusual current sign.

Recommended response:

1. Increase `V_max`.
2. Increase `n_points` or add a finer voltage region near the expected V_{oc}.
3. Inspect spatial profiles near the highest voltage.
4. Do not report FF or PCE from an unbracketed result.

15.2 Sentinel Or Zero FF/PCE

Zero FF or zero PCE does not always mean a physically dead solar cell. In SolarLab it can also mean the metric extraction refused to infer a maximum power point because V_{oc} was not bracketed. Always check `voc_bracketed` first.

15.3 TMM Or EQE Fails Because Optical Data Are Missing

EQE and EL require at least one layer with `optical_material`. If a stack only uses scalar `alpha`, it can run Beer-Lambert J-V but not wavelength-resolved experiments. Use a `*_tmm.yaml` preset or add verified n, k data.

15.4 YAML Values Look Numeric But Behave Strangely

YAML 1.1 can parse bare scientific notation such as `1e-9` as a string. SolarLab coerces numeric-looking fields in the config loader, but users should write `1.0e-9` for clarity and to avoid surprises in external tools.

15.5 Slow Or Stiff Runs

Thick absorbers, high fields, sharp interfaces, and ion-coupled transients can make Radau solves slow. Use a coarse grid while developing the configuration, then rerun final cases at publication settings. For CIGS or crystalline silicon, transient J-V may be much slower than equilibrium-style checks.

15.6 Unphysical Band Barriers

Large or inconsistent jumps in `chi` and `Eg` can create artificial heterojunction barriers. If a device has unexpectedly poor FF or current, plot the band diagram, check the transport-layer band offsets, and verify the contact velocities.

15.7 2D Results Differ From 1D

For a lateral-uniform 2D stack, differences from 1D should be small when ions are frozen consistently and the grids are comparable. If the difference is large, check:

- whether 1D ions were allowed to move while 2D ions were frozen;
- lateral boundary condition;
- `Nx`, `Ny_per_layer`, and the settling time (`settle_t`);
- whether a microstructure block was unintentionally active.

15.8 Frontend Job Appears To Hang

Long-running experiments use server-sent events. If the UI does not update, check that the backend was started from the SolarLab root with `--app-dir perovskite-sim`, then verify `/api/jobs/{id}/events` is reachable.

Chapter 16

Validation And Evidence

The Python evidence pass for this manual was executed on 2026-07-23 at commit `35e2f51` of the SolarLab repository (primary simulator tree `perovskite-sim/`), the same revision the manual text describes; an earlier full pass (2026-05-19, commit `43c81d7`) is superseded. The evidence should be read as implementation evidence: internal consistency of the equations, numerical coupling, backend/frontend interfaces, and benchmark envelopes at that repository state. It does not guarantee predictive accuracy for arbitrary material stacks.

The evidence in this chapter divides into three layers of increasing strength: (1) *unit/integration correctness* — the equations and interfaces compute what they claim; (2) *regression stability* — pinned baselines and acceptance envelopes that detect silent numerical drift; and (3) *external validation* — comparison against independent simulators or experiments. The figures below belong to layers 1 and 2: they are generated from test thresholds and pinned baselines rather than from independent data. Layer-3 evidence, the comparison of simulated metrics against the IonMonger and SCAPS reference outputs themselves, is documented in the benchmark and comparison campaigns separately.

Validation evidence executed before manual generation

Gate	Command	Result	Runtime
Python default suite	pytest	1101 passed	18.3 min
Python slow suite	pytest -m slow	99 passed, 1 xfail	80.9 min
Python validation suite	pytest -m validation	22 passed	7.4 min
Frontend build	npm run build	passed	6.3 min
Frontend unit tests	npm run test:run	371 passed	1.3 s

Figure 16.1: Validation gate summary

Gate	Command	Result	Time
Python default suite	`pytest`	1101 passed, 3 skipped, 1 xfailed	1098 s

Gate	Command	Result	Time
Python slow suite	`pytest -m slow`	99 passed, 1 xfailed (documented, see below), 4 skipped	4856 s
Python validation suite	`pytest -m validation`	22 passed	443 s
Frontend build	`npm run build`	passed (tsc clean; 86 modules)	380 s[a]
Frontend unit tests	`npm run test:run`	27 files, 371 tests passed	1.34 s[b]

The passing suites cover:

- core Poisson, recombination, ion, optics, contact, temperature, and trap modules;
- J-V, dark J-V, Suns-V_{oc}, EQE, EL, impedance, Mott-Schottky, TPV, degradation, tandem, and 2D workflows;
- backend API and SSE job dispatch;
- frontend result rendering and validation UI;
- IonMonger and Driftfusion-inspired benchmark envelopes;
- TMM Jsc baselines;
- photon-recycling voltage boost;
- 1D/2D lateral-uniform parity;
- 2D microstructure regression;
- physical trends for bandgap, thickness, mobility, dark ideality, and Suns-V_{oc} slope.

16.1 Validation Figures

The figures below summarize the validation evidence and slow-suite reference envelopes. They are derived from the tests and evidence files listed in the traceability matrix, rather than from independent simulation runs. The script used to generate them is `docs/manual/generate_manual_figures.py`.

The IonMonger benchmark gate uses `configs/ionmonger_benchmark.yaml` with `N_grid=40`, `n_points=20`, and `v_rate=5.0`. The plotted interval is the allowed tolerance around the pinned reverse-scan reference metrics: $V_{\text{oc}} = 1.1932\,V$, $J_{\text{sc}} = 231.70\,A\,m^{-2}$, $FF = 0.7774$, and $PCE = 0.2149$.

The TMM baseline gate constrains the n-i-p and p-i-n short-circuit currents to remain within the pinned optical-regression tolerance. This protects the transfer-matrix implementation from silent drift while allowing small numerical variation.

The photon-recycling regression requires the radiative-limit V_{oc} boost to remain inside the 40–100 mV literature window encoded by the test. The plotted marker gives the approximate MAPbI3 context value used for the regression.

The validation suite checks qualitative semiconductor-physics trends: current decreases with increasing band gap, mobility affects FF, dark J-V returns an ideality factor in the expected range, and Suns-V_{oc} produces a physically plausible slope.

The 2D slow-suite gates compare the lateral-uniform 2D solver against the 1D reference and verify that a single grain boundary produces a bounded voltage penalty rather than an uncontrolled numerical artifact.

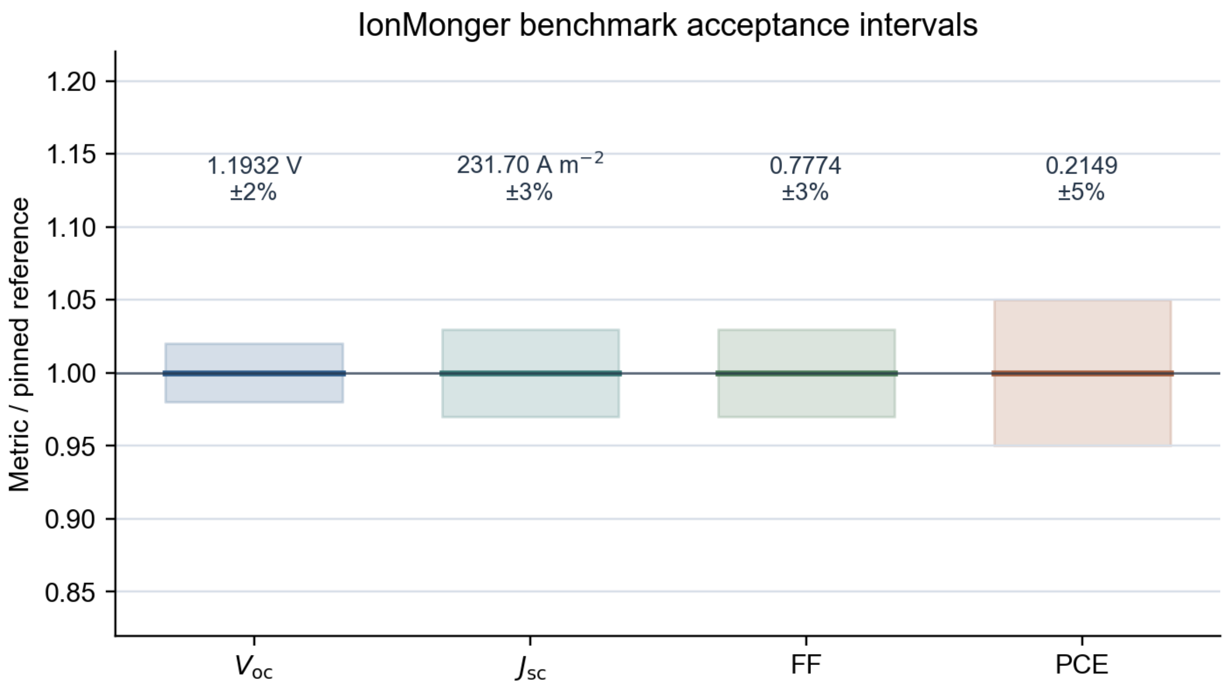

Figure 16.2: IonMonger reference metric envelope

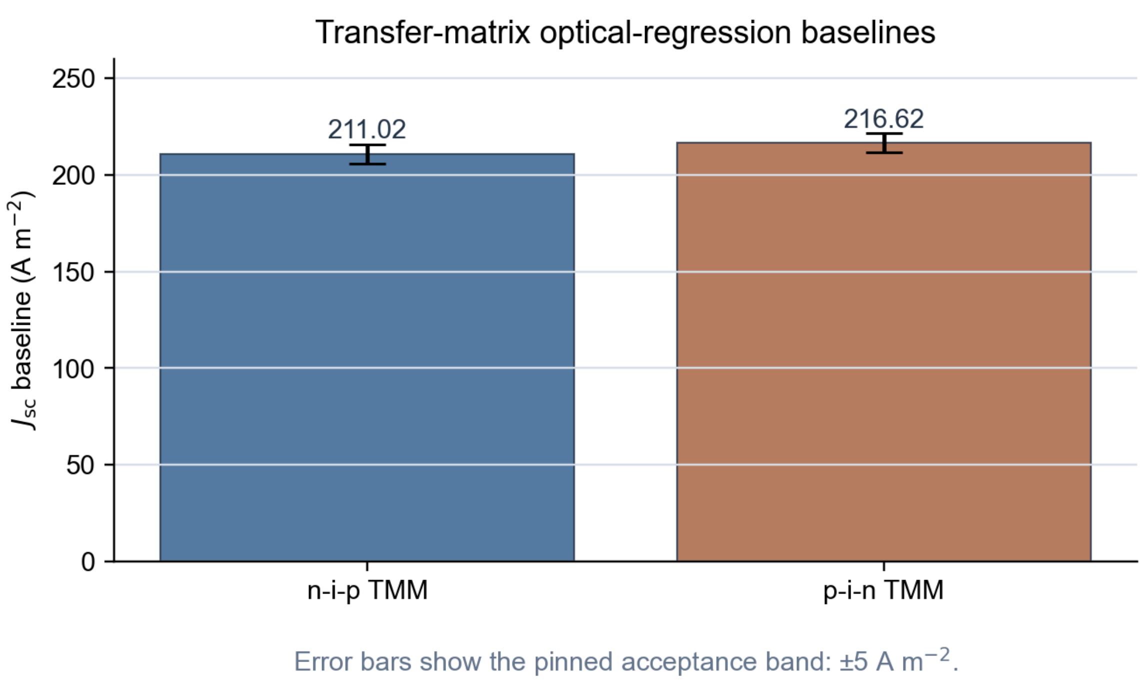

Figure 16.3: TMM optical baseline envelope

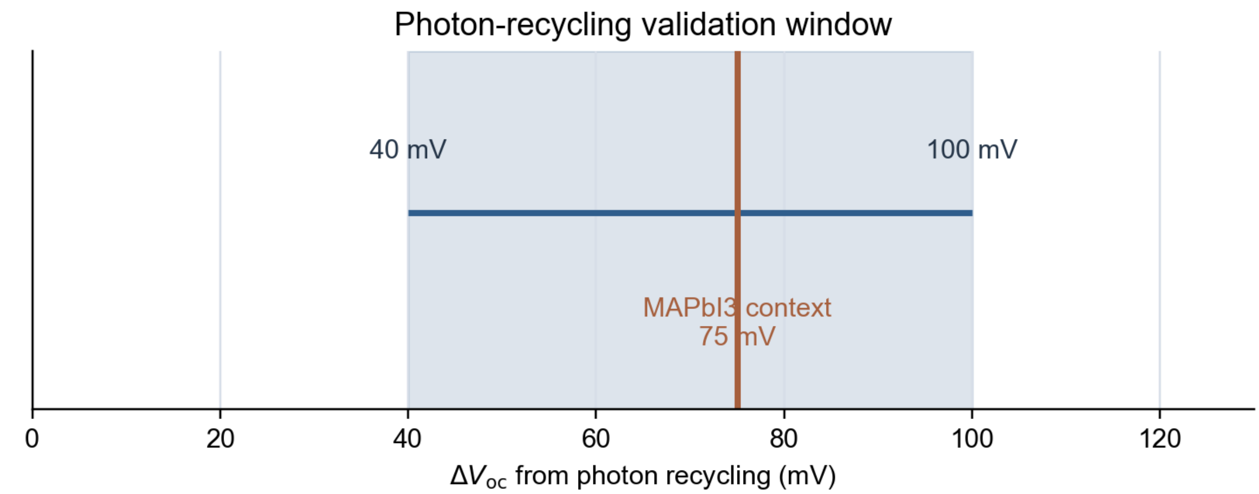

Figure 16.4: Photon-recycling validation window

Physics-trend validation matrix

Validation trend	Expected physical behavior	Evidence source
Bandgap	J_{sc} decreases as E_g increases	test_physical_trends.py
Thickness	V_{oc} changes 20-120 mV/decade	test_physical_trends.py
Mobility	FF responds to mobility change	test_physical_trends.py
Dark J-V	$1.0 \leq n_{id} \leq 2.5$	test_physical_trends.py
Suns-Voc	20-70 mV/decade slope	test_physical_trends.py

Figure 16.5: Physics trend validation matrix

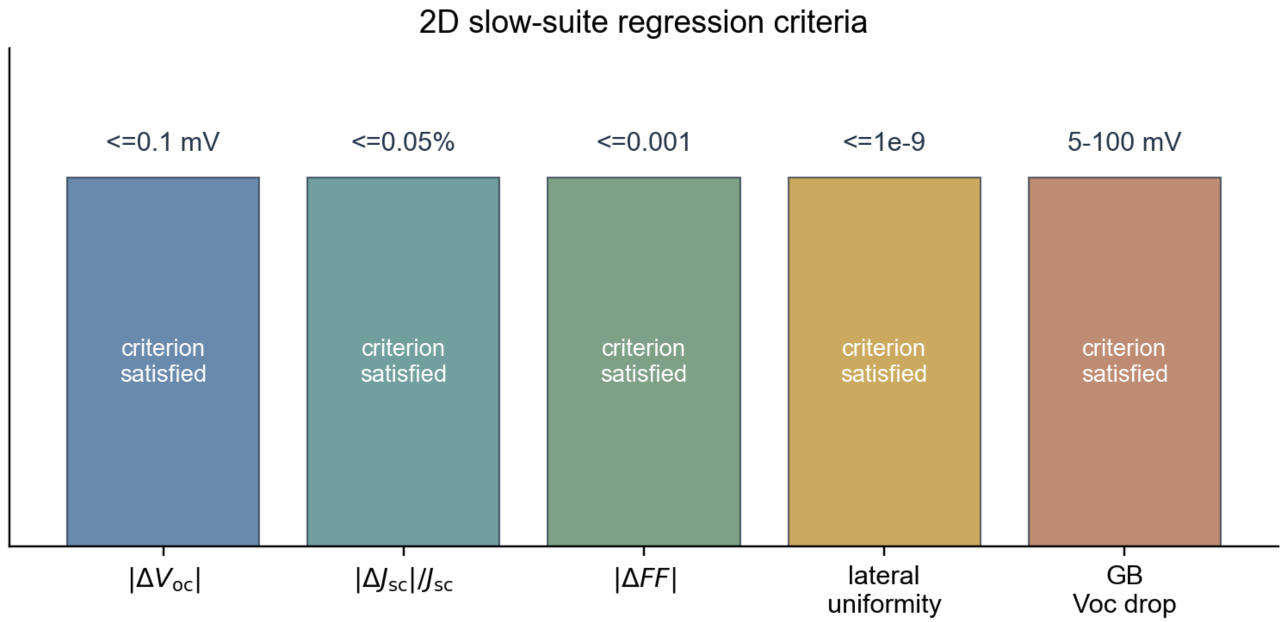

Figure 16.6: 2D validation and microstructure summary

16.2 Validation Traceability Matrix

Evidence item	Source test or file	What it defends
Default Python suite	`pytest`	broad unit/integration coverage for solver, models, backend, and experiments
Slow Python suite	`pytest -m slow`	expensive regression baselines, TMM, 2D parity, microstructure, benchmark envelopes
Validation suite	`pytest -m validation`	expected semiconductor-physics trends
IonMonger metric envelope	`perovskite-sim/tests/integration/test_voc_benchmark.py`	benchmark-scale J-V metrics and hysteresis bounds
TMM J_{sc} baselines	`perovskite-sim/tests/regression/test_tmm_baseline.py`	optical-stack regression stability
Photon-recycling boost	`perovskite-sim/tests/regression/test_photon_recycling_voc.py`	radiative-limit voltage boost sanity
2D parity	`perovskite-sim/tests/regression/test_twod_validation.py`	lateral-uniform 2D consistency with 1D
Grain-boundary voltage loss	`perovskite-sim/tests/regression/test_twod_microstructure.py`	bounded microstructure-induced voltage penalty
Frontend production build	`npm run build`	TypeScript/Vite production surface
Frontend unit tests	`npm run test:run`	UI result rendering, validation, progress, and workstation state

[a],Executed in an interactive terminal from the working tree; the wall time is dominated by cloud-synced-filesystem I/O, not compilation. [b],Executed at the same commit from a git checkout on local disk: the

cloud-sync file provider stalls the test runner's parallel file access when run in place; this reflects the cloud-synced filesystem rather than the test code.

The slow suite carries one expected failure (`xfail`):

- `test_grading_cigs_notch.py::test_cigs_back_grading_raises_voc_without_jsc_collapse` — a graded-CIGS back-surface-field regression that combines three conditions that jointly stress the coupled Radau solver: a 2 μm CIGS absorber (stiff; full transient J-V on 2 μm CIGS is outside the practical solver envelope, Chapter 11), a recombination-active Robin heterocontact (required by the back-surface-field premise, so it cannot be dropped), and a graded band-offset notch. Both the transient sweep (bisection exhaustion at the $V \approx 0.5$ V knee) and the direct steady-state Newton (unable to certify the $V = 0$ point) fail to reach V_{oc} on this configuration. The underlying back-surface-field physics is covered at the unit level; the test is marked `xfail` (non-strict, so it reports XPASS and flags itself if the CIGS solver envelope is later widened). This is a solver-envelope limitation, not a physics defect, present at the 2026-07-22 baseline commit.

An earlier autoloop-integration failure (`test_autoloop_boulder.py::test_boulder_sweep_real`, which pinned a gap proposal that no longer exists after the interface-channel calibration) was resolved in this revision by replacing the stale expectation with the sweep's structural invariant.

Observed warnings:

- `pytest_asyncio` future-default warning for fixture loop scope;
- `np.trapz` deprecation warnings in compatibility and conservation tests.

These warnings do not affect simulator correctness but should be resolved before a formal software release.

Chapter 17

Limitations And Best Practices

17.1 Model Assumptions

SolarLab is a numerical research simulator. Predictive accuracy for a material stack requires physically justified inputs and a validation envelope that covers the intended regime.

17.2 Known Formulation Limitations

Seven model-formulation limitations apply, each flagged at its point of use in Chapter 7:

- the *default* thermionic-emission bound is density-weighted and therefore an empirical interface-limited cap, not the dimensionally normalized Richardson-Dushman current (equilibrium safety follows from the cap construction). A dimensionally correct emission-velocity form is implemented as an opt-in (`te_physical_norm`), but it is off by default: on a density-of-states-bearing stack it raises the steady-state V_{oc} by about 0.3 V where the cap binds. It acts only on the bulk-node transport cap, not on the interface-plane-state formulation used for the SCAPS comparison (which is unaffected), so it does not disturb the calibrated parity result; it stays off by default because the paths it does affect lack an independent reference baseline with the physical cap active;
- the *default* ionic steric factor multiplies the full flux (drift and diffusion) and is applied per species, an empirical crowding regularization rather than the strict shared-site lattice-gas flux. The diffusion-only modified-PNP form is available as an opt-in (`ion_steric_diffusion_only`); it is numerically immaterial for the shipped presets (dilute ions, peak $P/P_{\text{lim}} \approx 0.011$, so V_{oc} shifts under 0.1 mV) and matters only near saturation;
- the interface-SRH non-negativity clamp (scoped to declared-defect interfaces) suppresses physical depletion-region generation there as the cost of the cross-carrier approximation; defect-free interfaces retain physical generation;
- the photon-recycling redistribution is spatially uniform, with no spatial or spectral reabsorption kernel;
- the Scharfetter-Gummel positivity property is a property of the *spatial* semi-discretization. The transient advances (n, p, P) with a general-purpose implicit integrator and applies no clipping or projection, so non-negativity is not guaranteed at every internal stage or accepted step. The direct steady-state driver does enforce it, by solving in logarithmic unknowns;
- the slow-ion operator splitting freezes the carriers over the macro-step and re-solves Poisson inside the ion RHS. That makes the field consistent with the instantaneous charge density but

does not hold the carriers quasi-stationary against the evolving ionic field, so it is a first-order splitting whose error grows with the macro-step and with the ion-induced band bending;
- the interface-plane thermal velocities and the density-of-states prefactor of the per-side interface-defect trap levels are fixed at their 300 K values, so the default-off interface formulations that use them are calibrated for room temperature and should not be combined with a temperature sweep without supplying that dependence explicitly.

Both optional forms are off by default: the physical thermionic-emission normalization changes bulk-node transport results that have no independent reference baseline, and the diffusion-only ion-steric form, though it does not perturb the current presets (sub-0.1 mV), is held pending validation against high-ion-density configurations.

17.3 Data And Optical Limits

TMM, EQE, and EL depend on tabulated n, k data. `load_nk` does not silently extrapolate outside the source wavelength range. Placeholder optical constants support workflow testing, but they do not support material-specific publication claims.

17.4 Numerical Limits

If a J-V sweep does not bracket V_{oc}, increase `V_max` or refine the voltage grid near the expected zero crossing. Sentinel zero FF or PCE should not be interpreted as physical failure until `voc_bracketed` has been checked.

Mott-Schottky fits can be unreliable for fully depleted thin absorbers.

Thick CIGS and crystalline silicon absorbers can be slow in transient mode.

17.5 2D Limits

The 2D solver freezes ions. It is suitable for lateral parity and microstructure studies under that approximation, but not for full dynamic ion migration in two dimensions.

17.6 Reporting Checklist

A reproducible SolarLab study should report:

- commit hash;
- device YAML or full parameter table;
- physics tier;
- grid settings;
- solver tolerances;
- optical data source;
- validation command results;
- whether V_{oc} was bracketed;
- known limitations relevant to the chosen experiment.

17.7 Publication Boundary Statement

A SolarLab figure is publication-grade only when all of the following are true:

- the device stack and all changed parameters are reported;
- optical constants and material parameters have provenance;
- the chosen physics tier is justified;
- the grid and solver settings are sufficient for the conclusion;
- validation gates relevant to the claimed regime have passed;
- limitations such as frozen 2D ions, placeholder optical data, or screening smoke settings are stated explicitly.

If any item is missing, the result may still be useful for debugging or hypothesis screening, but it should not be described as a calibrated material-specific prediction.

Chapter 18

SolarScale Screening Bridge

SolarLab can consume readiness-gated material records from SolarScale. This is a workflow layer, not the core solver.

Safe mappings include:

- dielectric constant to `eps_r`;
- electron and hole mobility to `mu_n` and `mu_p` after unit conversion;
- ion diffusion to `D_ion`;
- ion activation energy to `E_a_ion`;
- electron affinity (`electron_affinity_ev`) to the absorber `chi` — with the caution that this parameter directly sets the band alignment at both heterojunctions, so its provenance and uncertainty must accompany any screening result that depends on it.

Device-only unknowns should be swept rather than guessed:

- absorber thickness;
- SRH lifetime;
- trap density;
- interface recombination velocity;
- contact band alignment.

Smoke J-V results in screening workflows validate the process. They are not publication-grade device predictions unless production settings, provenance, and validation are also supplied.

Chapter 19

References

1. D. L. Scharfetter and H. K. Gummel, “Large-signal analysis of a silicon Read diode oscillator,” *IEEE Transactions on Electron Devices* **16**, 64–77 (1969). doi:10.1109/T-ED.1969.16566.
2. W. Shockley and W. T. Read, “Statistics of the recombinations of holes and electrons,” *Physical Review* **87**, 835–842 (1952). doi:10.1103/PhysRev.87.835.
3. R. N. Hall, “Electron-hole recombination in germanium,” *Physical Review* **87**, 387 (1952). doi:10.1103/PhysRev.87.387.
4. N. E. Courtier, J. M. Cave, J. M. Foster, A. B. Walker, and G. Richardson, “IonMonger: a free and fast planar perovskite solar cell simulator with coupled ion vacancy and charge carrier dynamics,” *Journal of Computational Electronics* **18**, 1435–1449 (2019). doi:10.1007/s10825-019-01396-2.
5. P. Calado, I. Gelmetti, B. Hilton, M. Azzouzi, J. Nelson, and P. R. F. Barnes, “Driftfusion: an open source code for simulating ordered semiconductor devices with mixed ionic-electronic conducting materials in one dimension,” *Journal of Computational Electronics* **21**, 960–991 (2022). doi:10.1007/s10825-021-01827-z.
6. M. Burgelman, P. Nollet, and S. Degrave, “Modelling polycrystalline semiconductor solar cells,” *Thin Solid Films* **361–362**, 527–532 (2000). doi:10.1016/S0040-6090(99)00825-1.
7. U. Rau, “Reciprocity relation between photovoltaic quantum efficiency and electroluminescent emission of solar cells,” *Physical Review B* **76**, 085303 (2007). doi:10.1103/PhysRevB.76.085303.
8. L. A. A. Pettersson, L. S. Roman, and O. Inganäs, “Modeling photocurrent action spectra of photovoltaic devices based on organic thin films,” *Journal of Applied Physics* **86**, 487–496 (1999). doi:10.1063/1.370757.
9. E. Yablonovitch, “Statistical ray optics,” *Journal of the Optical Society of America* **72**, 899–907 (1982). doi:10.1364/JOSA.72.000899.
10. D. M. Caughey and R. E. Thomas, “Carrier mobilities in silicon empirically related to doping and field,” *Proceedings of the IEEE* **55**, 2192–2193 (1967). doi:10.1109/PROC.1967.6123.
11. J. Frenkel, “On pre-breakdown phenomena in insulators and electronic semiconductors,” *Physical Review* **54**, 647–648 (1938). doi:10.1103/PhysRev.54.647.
12. M. A. Green, “Solar cell fill factors: General graph and empirical expressions,” *Solid-State Electronics* **24**, 788–789 (1981). doi:10.1016/0038-1101(81)90062-9.
13. J. Nelson, *The Physics of Solar Cells* (Imperial College Press, London, 2003).
14. P. Würfel, *Physics of Solar Cells: From Principles to New Concepts* (Wiley-VCH, Weinheim, 2005).
15. H. J. Snaith, “Perovskites: The emergence of a new era for low-cost, high-efficiency solar cells,” *Journal of Physical Chemistry Letters* **4**, 3623–3630 (2013). doi:10.1021/jz4020162.

16. L. M. Pazos-Outón et al., “Photon recycling in lead iodide perovskite solar cells,” *Science* **351**, 1430–1433 (2016). doi:10.1126/science.aaf1168.
17. S. Selberherr, *Analysis and Simulation of Semiconductor Devices* (Springer, Vienna, 1984).
18. E. Hairer and G. Wanner, *Solving Ordinary Differential Equations II: Stiff and Differential-Algebraic Problems*, 2nd ed. (Springer, Berlin, 1996).
19. G. F. Burkhard, E. T. Hoke, and M. D. McGehee, “Accounting for interference, scattering, and electrode absorption to make accurate internal quantum efficiency measurements in organic and other thin solar cells,” *Advanced Materials* **22**, 3293–3297 (2010). doi:10.1002/adma.201000883.
20. F. A. Padovani and R. Stratton, “Field and thermionic-field emission in Schottky barriers,” *Solid-State Electronics* **9**, 695–707 (1966). doi:10.1016/0038-1101(66)90097-9.
21. Y. P. Varshni, “Temperature dependence of the energy gap in semiconductors,” *Physica* **34**, 149–154 (1967). doi:10.1016/0031-8914(67)90062-6.
22. M. S. Kilic, M. Z. Bazant, and A. Ajdari, “Steric effects in the dynamics of electrolytes at large applied voltages. II. Modified Poisson-Nernst-Planck equations,” *Physical Review E* **75**, 021503 (2007). doi:10.1103/PhysRevE.75.021503.
23. H. Pauwels and G. Vanhoutte, “The influence of interface state and energy barriers on the efficiency of heterojunction solar cells,” *Journal of Physics D: Applied Physics* **11**, 649–667 (1978). doi:10.1088/0022-3727/11/5/009.
24. M. Burgelman and J. Marlein, “Analysis of graded band gap solar cells with SCAPS,” *Proceedings of the 23rd European Photovoltaic Solar Energy Conference*, 2151–2155 (2008).

Chapter 20

Appendix A: Manual Source Trail

The manual was drafted from:

- `docs/solarlab_manual_source_dossier.md`
- `docs/manual/valEvidence260723.md` (current evidence pass)
- `docs/manual/valEvidence260519.md` (superseded 2026-05-19 pass)
- `perovskite-sim/AGENTS.md`
- `README.md`
- `perovskite-sim/README.md`
- `perovskite-sim/backend/README.md`
- `perovskite-sim/perovskite_sim/models/`
- `perovskite-sim/perovskite_sim/physics/`
- `perovskite-sim/perovskite_sim/solver/`
- `perovskite-sim/perovskite_sim/experiments/`
- `perovskite-sim/perovskite_sim/twod/`
- `perovskite-sim/backend/main.py`
- `perovskite-sim/frontend/src/types.ts`

This PDF is intended to be defensible as a project manual. For a journal supplement, add parameter provenance for every material value and verify the full bibliography format against the target journal style.

Chapter 21

Appendix B: Figure Source Trail

Figure	Source	Reproducibility note
Architecture flow	`docs/manual/generate_manual_figures.py`	Diagrammatic summary of repo architecture described in `perovskite-sim/AGENTS.md`.
Validation gate summary	`docs/manual/valEvidence260723.md`	Uses pass counts and runtimes from the completed 2026-07-23 validation pass.
IonMonger reference metrics	`perovskite-sim/tests/integration/test_voc_benchmark.py`	Shows pinned benchmark metrics and tolerances, not a new simulation run.
TMM J_{sc} baselines	`perovskite-sim/tests/regression/test_tmm_baseline.py`	Shows pinned n-i-p and p-i-n TMM baselines with ±5 $A\,m^{-2}$ tolerance.
Photon-recycling window	`perovskite-sim/tests/regression/test_photon_recycling_voc.py`	Shows the acceptance window used by the slow regression gate.
Physics trend matrix	`perovskite-sim/tests/validation/test_physical_trends.py`	Summarizes physical trend assertions represented by the validation suite.
2D validation summary	`perovskite-sim/tests/regression/test_twod_validation.py`, `perovskite-sim/tests/regression/test_twod_microstructure.py`	Summarizes parity and grain-boundary gates represented by the slow suite.

Chapter 22

Appendix C: Quick Glossary

Term	Meaning
AM1.5G	Standard terrestrial solar spectrum used for photovoltaic testing.
Beer-Lambert	Scalar absorption law using an absorption coefficient α.
Bernoulli function	$B(\xi) = \xi/(e^{\xi} - 1)$; the weighting function of the Scharfetter-Gummel flux.
Detailed balance	Requirement that every generation/recombination channel vanishes exactly at thermodynamic equilibrium.
Displacement current	Capacitive current $\varepsilon_0\varepsilon_r\,\partial E/\partial t$ included in the terminal current.
Drift-diffusion	Continuum transport model combining diffusion from concentration gradients and drift from electric fields.
FF	Fill factor, the ratio between maximum power and $V_{\mathrm{oc}}J_{\mathrm{sc}}$.
Hysteresis	Difference between forward and reverse J-V scans caused by state memory, often ionic in perovskites.
Ideality factor	Slope parameter n_{id} of the dark diode characteristic.
Method of lines	Numerical method that discretizes space first and integrates the resulting ODE system in time.
PCE	Power-conversion efficiency.
Photon recycling	Reabsorption of internally emitted photons, reducing the net radiative loss.
Quasi-Fermi level	Electrochemical potential of one carrier population out of equilibrium; flat quasi-Fermi levels mean zero current.
Radau IIA	L-stable fully implicit Runge-Kutta method used for the stiff time integration.

Term	Meaning
Richardson constant	Prefactor A^* of the thermionic-emission flux across a band offset.
SG flux	Scharfetter-Gummel flux discretization, used for stable drift-dominated carrier transport.
SRH	Shockley-Read-Hall trap-assisted recombination.
Steric limit	Finite site density P_{lim} bounding mobile-ion accumulation.
Thermionic emission	Interface-limited carrier transport over a heterojunction band offset.
TMM	Transfer-matrix method for wavelength-resolved multilayer optics.
J_{sc}	Short-circuit current density.
V_{oc}	Open-circuit voltage.